Sovereign Trading Platform: Architecture, Engineering & Verification Report

Field	Value
Project	Sovereign Algorithmic Trading Platform (SOVEREIGN)
Scope	Distributed multi-broker trading, quantitative analytics and storage infrastructure
Target environment	Linux x86_64 (Debian 13 / Ubuntu LTS), Proxmox VE 8+ / LXC containers
Document type	Architecture specification, code walkthrough, verification report
Date of record	October 2026

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Executive Summary

The **Sovereign Trading Platform** is a high-performance, keyless-first algorithmic research and multi-broker execution system. It covers the full quantitative lifecycle: tick ingestion, order book microstructure analysis, frame backtesting, deterministic paper execution, and gated live broker routing.

This document is the canonical architecture specification and verification report. It covers engine principles, data storage invariants, cryptographic paper ledgers, strategy taxonomy, container deployment topology, and the testing pyramid that governs the codebase.

Chapter 1. Introduction, Problem Statement & Market Invariants

1.1 Problem Statement

Quantitative trading systems operate under strict latency, memory and consistency constraints. Across equity exchanges (NASDAQ, NYSE), crypto derivative venues (Deribit, Gate.io) and binary prediction markets (Polymarket CLOB), engines commonly fail in four ways:

#	Failure mode	Description
1	**Memory bloat and GC jitter (the JSON pitfall)**	Storing tick and candle history as JSON objects adds roughly 10x–20x overhead over the raw numeric payload. In V8, multi-megabyte allocations cause stop-the-world GC pauses (15–200 ms) and OOM crashes in 512 MB–1 GB containers.
2	**Backtest-to-live divergence**	Strategies overfit to historical mid-prices. Live alpha disappears due to spread crossing, adverse selection, market impact, queue latency and non-linear fees.
3	**Over-privileged daemons and credential sprawl**	Combining dashboards, research bots and live execution in one process exposes API keys and trading PINs to untrusted perimeters.
4	**Shallow mock verification**	Tests inject hardcoded mock stubs that bypass real codecs, deserializers, disk buffers and network edge cases. API format drift then crashes live loops.

Sovereign addresses these with native binary storage, strict privilege boundaries, zero-key local reproducibility and low-allocation compute engines.

1.2 Implemented Functionality

Domain	Capabilities
a) Market ingestion and bitemporal data	Multi-provider realtime and historical backfill (Alpaca Data v2, Polymarket CLOB REST/WS, Gate.io WebSocket, Yahoo Finance, FRED). Monotonic binary compaction into packed 48-byte records (`.bin`). Atomic staging writes (`.tmp` , PID-isolated) before canonical rename. Anomaly quarantine in `storage/data/quarantine/` .
b) Broker execution gateways	Unified abstraction over Alpaca Markets, Polymarket CLOB, Gate.io and MetaTrader 5. Read-only public monitoring with zero credentials. EIP-712 signing for prediction contracts. Rate-limited, circuit-breaker-guarded REST/WS pools.
c) Cryptographic paper ledger	SHA-256 chained transaction log. Depth-aware matching engine with VWAP slippage. Multi-strategy virtual sub-positions on overlapping symbols. Binary contract settlement and receivable locking.
d) Quantitative analytics	Indicators: RSI, EMA, SMA, ATR, Bollinger Bands, Volume Oscillator. 200-bar lookback fastpath for live loops. C++20 SIMD frame backtester. Combinatorial probability arbitrage discovery.
e) Risk and operational defense	Multi-tier pre-trade risk (position ceilings, daily loss, leverage). File-based and API kill-switch with mass cancellation. Fail-closed root filesystem monitor. Per-daemon heartbeat with TTL tracking.
f) Operator interfaces	Ink/Node.js terminal UI. Private authenticated loopback Express API (`127.0.0.1:8787`). React 19 + Vite dashboard. Diagnostics CLI (`sovereign system-doctor`).

1.3 Market Invariants & Microstructure Rules

All data entering the system must satisfy explicit invariants.

Strict monotonicity of event time

```
T_event[i] < T_event[i+1]   for sequential regular bars on symbol S
```

Out-of-order ticks are intercepted by bitemporal sorting buffers before index commitment.

Non-crossed book

```
Bid_top < Ask_top   =>   Spread = Ask_top - Bid_top > 0.00
```

If `Bid >= Ask` , the book is classified as crossed, halted from execution and quarantined.

Positive price and non-negative volume

```
Open, High, Low, Close > 0.00
Low <= min(Open, Close) <= max(Open, Close) <= High
Volume >= 0.00
```

Polymarket binary complementarity

For complementary outcomes resolving to 1.00 USD:

```
P(Yes) + P(No) ≈ 1.00
```

When the spread exceeds 0.10 USD (`Ask_Yes - Bid_Yes > 0.10`), canonical reference pricing switches away from the illiquid mid-quote:

```
P_ref(Yes) = lastTradePrice   when lastTradePrice is finite and positive
```

The complementary price `1.00 - Ask_No` remains a separate NegRisk consistency bound; it is not the production reference-price fallback. The executable owner is `calculatePolymarketReferencePrice` in `shared/lib/strategy/polymarket_clob_model.js` .

Paper ledger balance conservation

```
Equity_T = Cash_T + sum(Quantity_k * MarkPrice_k) + Receivables_pending
```

No transaction may create or destroy capital outside realized PnL or audited adjustments.

Figure 1.1: Multi-Venue Ingress & Invariant Validation Pipeline

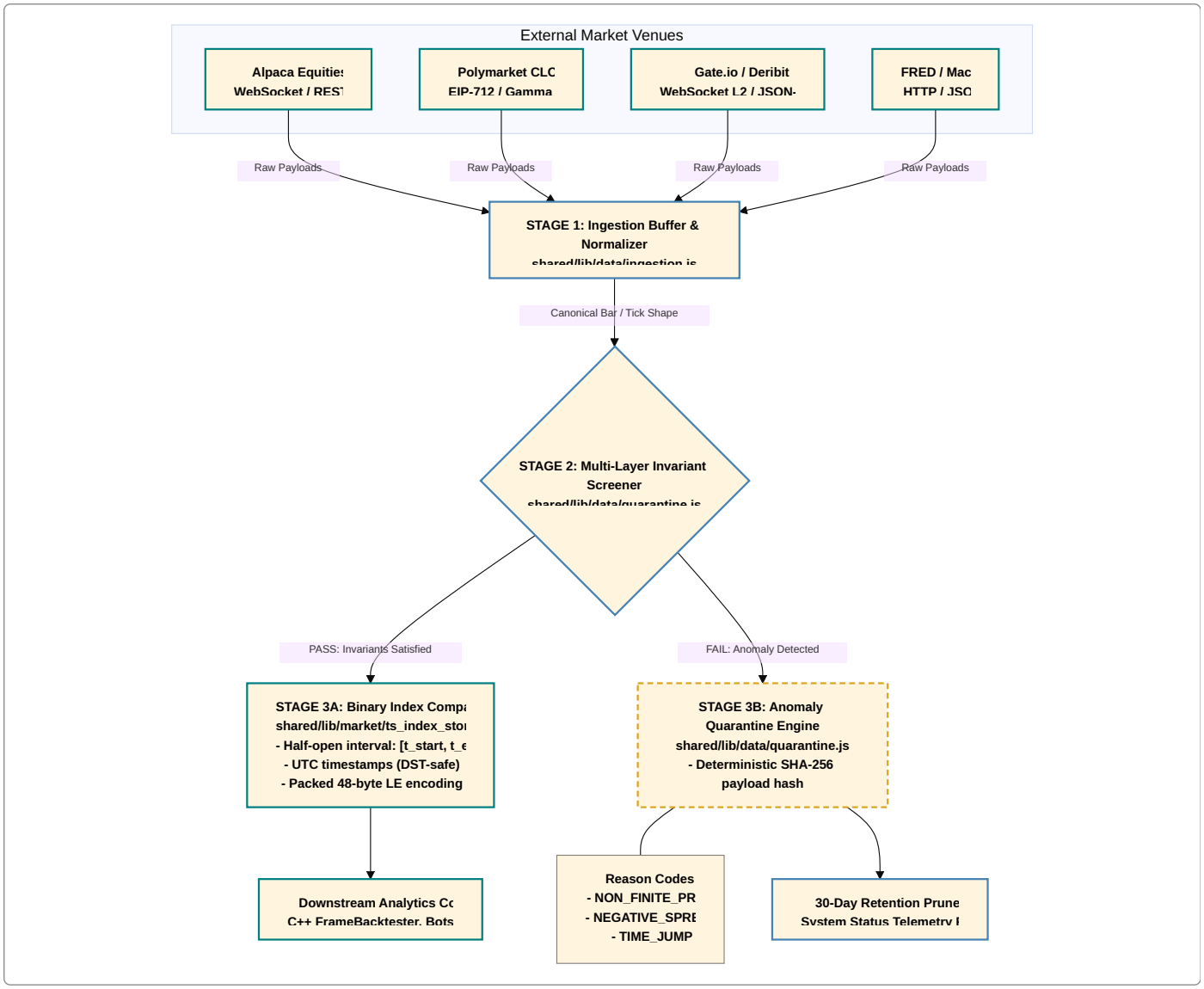
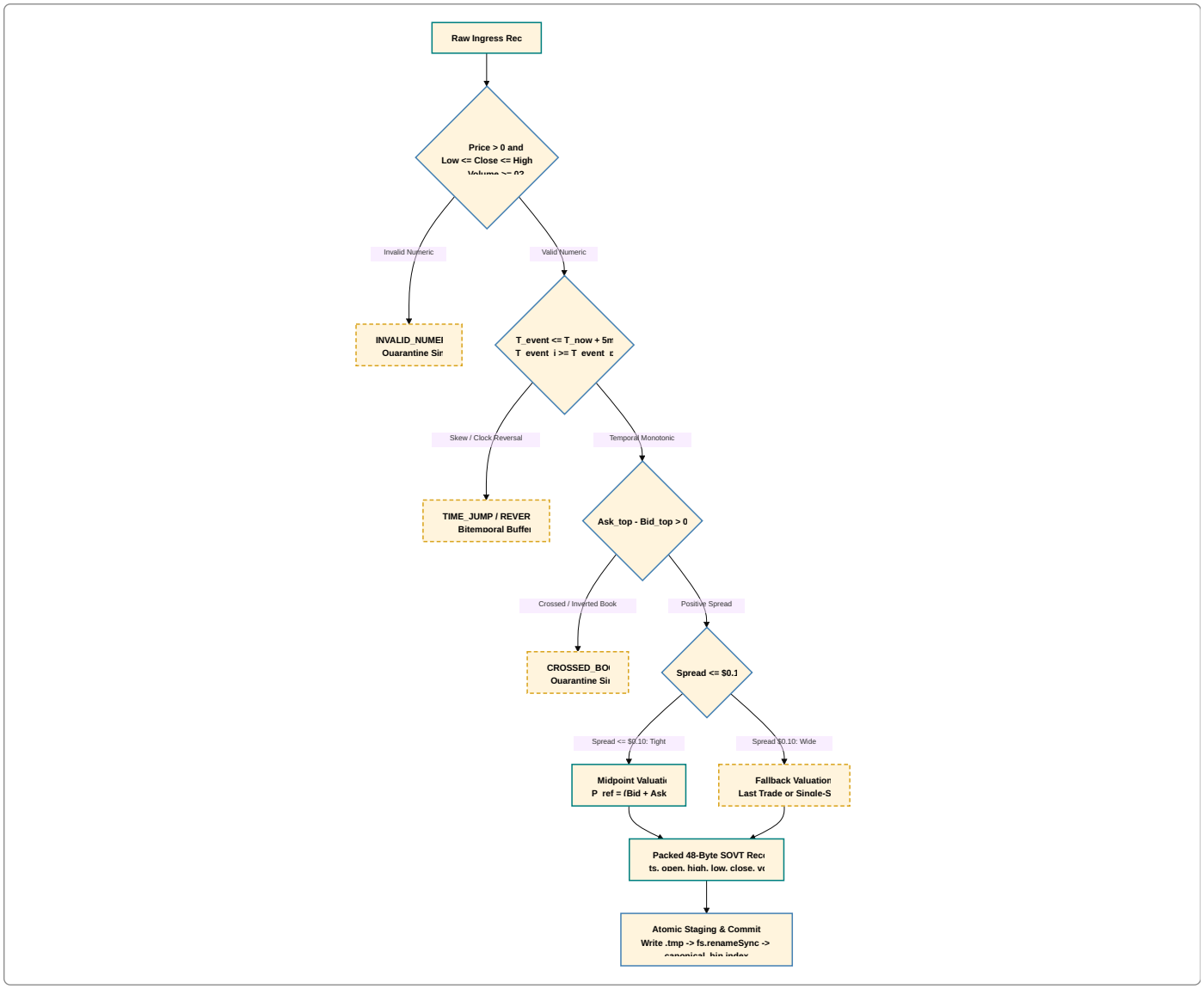


Figure 1.2: Microstructure Invariant State Transition Topology



1.4 State Machine Enums

State machine	Values	Transitions and rules
Service heartbeat	`starting`, `running`, `healthy`, `degraded`, `stopped`, `unavailable`	Monitored by `service_heartbeat.js`. Becomes `degraded` on disk threshold or broker timeout. TTL expiry marks a service `unavailable`.
Order lifecycle	`pending_submit`, `submitted`, `partially_filled`, `filled`, `cancelled`, `rejected`, `expired`	Verified via broker fill frames or paper ledger execution. Order and filled quantity are conserved across partial fills.
Broker session auth	`unauthenticated`, `authenticated`, `simulated_zero_key`, `circuit_broken`	Validated by adapters. Zero-key mode isolates execution to paper simulation.
Quarantine code	`NON_FINITE_PRICE`, `NEGATIVE_SPREAD`, `TIME_JUMP`, `MALFORMED_ROW`	Assigned from `QUARANTINE_REASONS` in `quarantine.js`. The raw payload is written to an isolated JSONL audit record.

1.5 System-Wide Requirements

1. **Zero-key deterministic development:** the whole codebase (ingestion regressions, indicators, backtesting, REST/WS endpoints) runs locally without external keys.
2. **Fail-closed security boundaries:** unauthenticated, expired or malformed requests return HTTP 401/403 without leaking stack traces or paths.
3. **Atomic disk operations:** no direct writes to live files (`.bin`, `.json`). Write to a process-isolated `.tmp` file, then commit via POSIX atomic rename.
4. **Decoupled business logic:** models, risk policies and storage codecs do not depend on CLI, HTTP or UI layers.
5. **No untested code:** every algorithm, fix or config change ships with an exercising test. No shallow mocks.

Chapter 2. Development Environment, Tooling & Polyglot Architecture

2.1 Environment & Tooling

Item	Specification
OS	Linux 6.12.x x86_64 (Debian 13 / Ubuntu LTS); bare-metal host `hpdesk` and Proxmox VE container CT 114
Node.js	v20.x+ LTS (ESM/CommonJS interop via TypeScript 5.x)
C++	ISO C++20 (`-std=c++20`)
Go	1.22+ (streaming daemons)
Toolchain	GCC 13+ / Clang 17+ with `-O3 -march=native -Wall -Wextra -Werror -fPIC` ; CMake 3.15+ with CTest
Editors	VS Code and Neovim with clangd, tsserver, GDB/LLDB

Build and test commands

Command	Purpose
`npm run setup:dev`	Initialize workspace, seed fixtures, install dependencies
`npm run native:build`	Compile the C++20 engine via CMake
`npm run test:core`	Native CTest suite (34/34 passing)
`npm run test:structure`	Structural contracts, skill integrity, repo hygiene
`npm run test:data`	Ingestion regression and indicator data flow
`npm run test:api`	Private Express API security and contract tests
`npm test`	Full suite

2.2 Standard Library Inventory

Node.js (`node:*`)

Module	Use
`fs` , `fs/promises`	POSIX file I/O, descriptor streaming, `fs.statfsSync` for disk checks
`path`	Path resolution, boundary checks, traversal shields
`crypto`	SHA-256 and HMAC-SHA256 for ledger chains; `timingSafeEqual` for tokens
`child_process`	Spawning native binaries and diagnostics
`stream` , `stream/promises`	Backup tarball and cache pruning pipelines
`test` , `assert/strict`	Zero-dependency test runner and strict assertions
`v8`	`getHeapStatistics` for anti-OOM diagnostics
`os`	CPU topology, load averages, memory accounting
`net` , `http`	Loopback networking, IPv4/IPv6 CIDR classification, socket timeouts

C++20 (`std::`)

Header	Use
`<iostream>`, `<fstream>`, `<sstream>`	Binary file I/O for 48-byte records
`<vector>`, `<array>`, `<span>`	Contiguous buffers for vectorized indicators
`<algorithm>`, `<numeric>`	Sorting, partitioning, reductions
`<chrono>`	Steady-clock timing, ns conversions
`<thread>`, `<atomic>`, `<mutex>`	Concurrent replay and lock-free evaluation
`<bit>`	Endianness handling (`std::endian`)
`<concepts>`	Compile-time constraints on bar types and strategy interfaces

Figure 2.1: Polyglot Execution & IPC Architecture

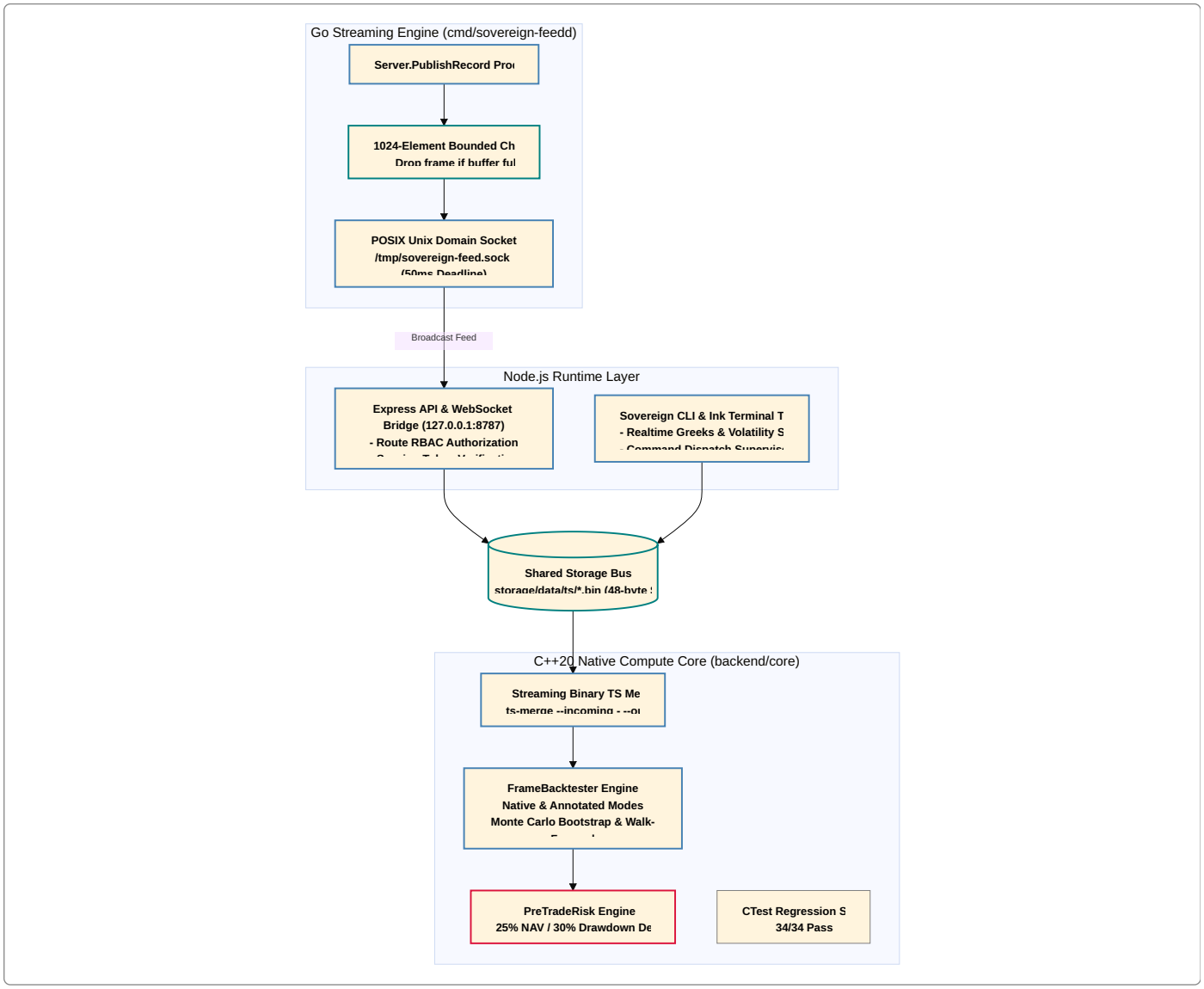
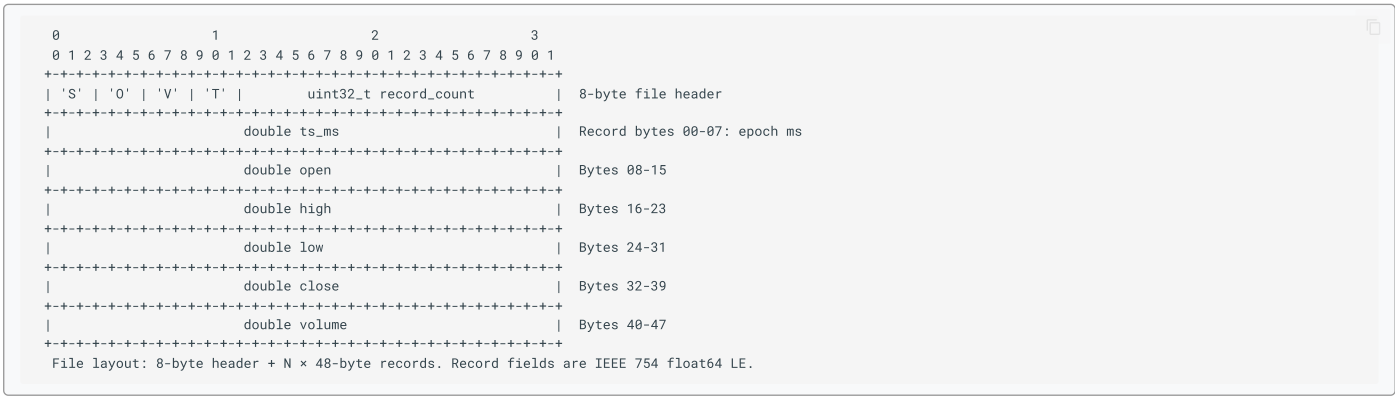


Figure 2.2: Packed 48-Byte `RawTsRecord` Memory Layout



The canonical format is owned by `TS_HEADER_BYTES` and `TS_RECORD_BYTES` in `shared/lib/market/ts_index_storage.js`: four ASCII magic bytes (`SOVT`), a `uint32LE` record count, then six `float64LE` values per record.

2.3 Custom Code Inventory (by Layer)

Layer 1: Shared runtime, invariants & storage

`shared/lib/runtime/host_maintenance.js`: health inspection, root filesystem fail-closed checks, host backups, manifest verification, retention pruning.

Function	Purpose
`diskUsage(targetPath)`	Queries filesystem capacity (total, free, free %) via `fs.statfsSync`.
`checkRootDiskSpace(rootPath, options)`	Raises `ENOSPC` if free space < 1 GiB or free % < 5%.
`newestCanonicalBinMtime(root)`	Finds the most recent `.bin` modification time.
`probeRunner(statusPath, now, maxAgeMs)`	Checks that daemons are running with fresh timestamps.
`probeHost(options)`	Aggregates data freshness, disk capacity and runner status.
`completedBackups(backupRoot, sourceRoot)`	Lists valid backups by presence of `manifest.json`.
`validateRetentionPolicy(maxAgeMs, maxCount)`	Validates retention bounds.
`pruneHostBackups(options)`	Prunes archives past age/count limits, preserving newest and protected paths.
`createHostBackup(options)`	Copies state into a `.tmp` dir, hashes files (SHA-256), writes `manifest.json`, commits via atomic rename.

`shared/lib/market/ts_index_storage.js`: packed 48-byte candle serialization, seeking and deserialization.

Function	Purpose
`readLatestTsRecord(filePath)`	O(1) tail read: seeks `filesize - 48`.
`readTsIndex(filePath, options)`	Reads records into candle arrays with optional start/end slicing.
`writeTsIndex(filePath, records)`	Serializes to packed little-endian buffer; commits atomically via `.tmp`.

`shared/lib/data/quarantine.js`: detects, classifies and sequesters malformed frames.

Function	Purpose
`validateIngestRecord(record, family)`	Checks finite positive price, `low <= close <= high`, non-negative volume, forward time bounds.
`quarantineRecord(record, reasonCode, errorMessage, options)`	Appends to `storage/data/quarantine/{family}_{YYYY-MM-DD}.jsonl` with SHA-256 digest.
`getQuarantineMetrics(options)`	Returns `total_quarantined`, `file_count`, `by_reason`, `by_family`, `latest_anomaly`.
`pruneQuarantineRecords(daysToKeep, options)`	Unlinks logs older than `daysToKeep` (default 30); returns `{ pruned_files, freed_bytes }`.

`shared/lib/data/db_pruning.js`

Function	Purpose
`runMaintenance(daysToKeep, archiveDir, options)`	Coordinated retention across Supabase tables (`macro_observations`, `orders`) and quarantine logs.

`shared/lib/runtime/service_heartbeat.js`

Function	Purpose
`writeServiceHeartbeat(service, patch)`	Atomically updates `storage/data/cache/service_heartbeats/{service}.json`.
`readServiceHeartbeat(service, options)`	Reads heartbeat; flags missing, invalid or expired states.
`errorCode(message)`	Maps errors to `storage_unavailable`, `authentication_failed`, `upstream_timeout_or_rate_limited`, `permission_denied`.

Layer 2: Indicators & strategy models

`shared/lib/market/indicators.js`

Function	Purpose
`rsi(closes, period)`	Wilder-style relative strength index series.
`ema(data, period)`	Exponential moving average series.
`sma(data, period)`	Rolling simple moving average series.
`atr(bars, period)`	Average true range over canonical OHLC bars.
`bollingerBandsSeries(closes, period, multiplier)`	Upper, middle and lower Bollinger band series.

`shared/lib/strategy/polymarket_clob_model.js`

Function	Purpose
`calculatePolymarketReferencePrice(bestBid, bestAsk, lastTradePrice)`	Midpoint pricing for tight books and last-trade fallback when spread > 0.10 USD.
`calculateDepthImbalance(bids, asks, depth, options)`	Harmonic depth imbalance, micro-price and reference-price metadata.
`forecastBinaryProbability(pMid, dir5, ofiZ, weights)`	Bounded logit probability drift with pin-risk protection.
`calculateMarketFillCost(asks, targetShares)`	Depth-walking VWAP and remaining-liquidity calculation.
`detectSumToOneArbitrage(outcomes, options)`	Multi-outcome complete-set cost and edge evaluation.

`shared/lib/market/multi_grain_rollup.js`

Function	Purpose
`deriveMultiGrainInMemory(oneMinBars, targetGrain)`	Rolls 1m bars into 5m/15m/1h/1d candles.
`alignGrainBoundary(timestamp, grainMinutes)`	Rounds epoch ms down to interval boundaries.

Layer 3: Broker gateways & paper ledger

`backend/gateway/src/polymarket/paper_ledger.ts`

Function	Purpose
`buildEvent(input, sequence, priorChecksum)`	Normalizes a ledger event and attaches its forward checksum.
`appendLedgerEvents(storageDir, inputs, options)`	Appends idempotent events under an exclusive lock and refreshes the projection snapshot.
`readLedger(storageDir)`	Reads and verifies the append-only event chain.
`replayLedger(events, initialBalance, options)`	Projects virtual balance, settled cash, receivables, positions and fills at `asOf`.
`initializeLedger(storageDir, startingBalance, options)`	Creates the deterministic genesis event.
`loadLedgerProjection(storageDir, startingBalance, options)`	Loads verified events and returns their time-grounded projection.

`backend/gateway/src/adapters/`

Adapter	Function	Purpose
`alpaca_adapter.ts`	`submitOrder(params)`	Limit/market/stop orders, fractional shares.
	`getPositions()`	Open positions with entry price and unrealized PnL.
	`streamBars(symbols, timeframe, onBar)`	Resilient WebSocket bar stream.
	`cancelAllOrders()`	Kill-switch mass cancel.
`polymarket_adapter.ts`	`buildEip712Order(params)`	Builds and signs typed order payloads.
	`postClobOrder(signedOrder)`	Submits to the CLOB REST endpoint.
	`subscribeOrderBook(tokenId, onUpdate)`	Level-2 WebSocket stream.
	`cancelOrder(orderId)`	Cancels resting limit orders.
`gate_io_adapter.ts`	`subscribeSpotTicker(symbol, onTicker)`	Public ticker stream, no keys.
	`fetchOrderBook(symbol, depth)`	Public L2 snapshot for parity validation.

Layer 4: Native analytics engine

File	Function	Purpose
`backend/core/src/backtest/frame_backtester.{cpp,hpp}`	`sovereign::FrameBacktester::run(bars, config)`	Frame backtest over contiguous records.
	`computeMetrics()`	Sharpe, Sortino, max drawdown, win rate and profit factor.
	`exportJson(outputStream)`	Serializes the report.
`backend/core/src/data/binary_ts_merger.{cpp,hpp}`	`sovereign::BinaryTsMerger::merge(baseFile, deltaFile, outputFile)`	O(N+M) two-pointer merge over 48-byte records with deterministic duplicate resolution.
	`validateSorted(filePath)`	Confirms monotonic timestamps across a file.

Layer 5: Operational daemons, deployment & API

`backend/scripts/ops/host_backup.js` (wrapper: `scripts/ops/host_backup.js`)

Function	Purpose
`parseHostBackupOptions(args, env)`	Maps CLI args and `HOST_BACKUP_ROOT`, interval, retention.
`resolveInitialBackupDueAt(options)`	Next run from heartbeat and manifest timestamps.
`runBackupAttempt(options)`	One attempt with fail-closed error classification.
`runHostBackupLoop(options)`	Watch loop; startup disk safety check; heartbeat updates.
`exitCodeForResult(result)`	`0` success, `1` fatal, `3` retention-only warning.

`scripts/deploy/deploy_proxmox.sh`

Mechanism	Purpose
`flock -n 9`	Prevents concurrent deploys.
`check_root_disk`	Requires ≥ 1 GB and ≥ 5% free root space.
`create_pre_deploy_snapshot`	Tarball of persistent storage in `/data/backups/`.
`rollback`	Restores previous image and storage snapshot if health checks fail.

`backend/api/app.js`, **`backend/api/server/services/access_control.js`**: loopback Express API (`127.0.0.1:8787`).

Function	Purpose
`constantTimeEqual(left, right)`	Timing-safe comparison for equal-length non-empty credentials.
`resolvePrincipal(req, options)`	Resolves service, human and approved loopback principals.
`resolveSocketPrincipal(socket, options)`	Applies the same principal contract to WebSocket handshakes.
`isLocalGuestAllowed(env)`	Gates zero-key loopback operator access by environment policy.

2.4 Language Responsibilities

Runtime	Responsibility
Node.js / TypeScript	Event-loop I/O, WebSocket streaming, Express server, TUI
C++20	Rolling-window math, SIMD frame backtesting, binary TS merging
Go 1.22+	Binary packet decode and UDS streaming daemons

Chapter 3. Ingestion, Binary TS Storage & Invariant Quarantine

3.1 Packed 48-Byte Binary TS Format

Path: `storage/data/ts/{SYMBOL}_{TIMEFRAME}.bin`

```
File 00..03:  [ ASCII      magic     ]  `SOVT`
File 04..07:  [ uint32_le  count     ]  record count
Record 00..07:[ float64_le timestamp ]  Unix epoch, milliseconds
Record 08..15:[ float64_le open      ]
Record 16..23:[ float64_le high      ]
Record 24..31:[ float64_le low       ]
Record 32..39:[ float64_le close     ]
Record 40..47:[ float64_le volume    ]
```

Each record is exactly 48 bytes after the 8-byte file header. One year of 1-minute bars (about 525,600 candles) occupies about 25.2 MB before filesystem overhead. Load latency is environment-dependent and requires a dated benchmark command before it can be stated as a guarantee.

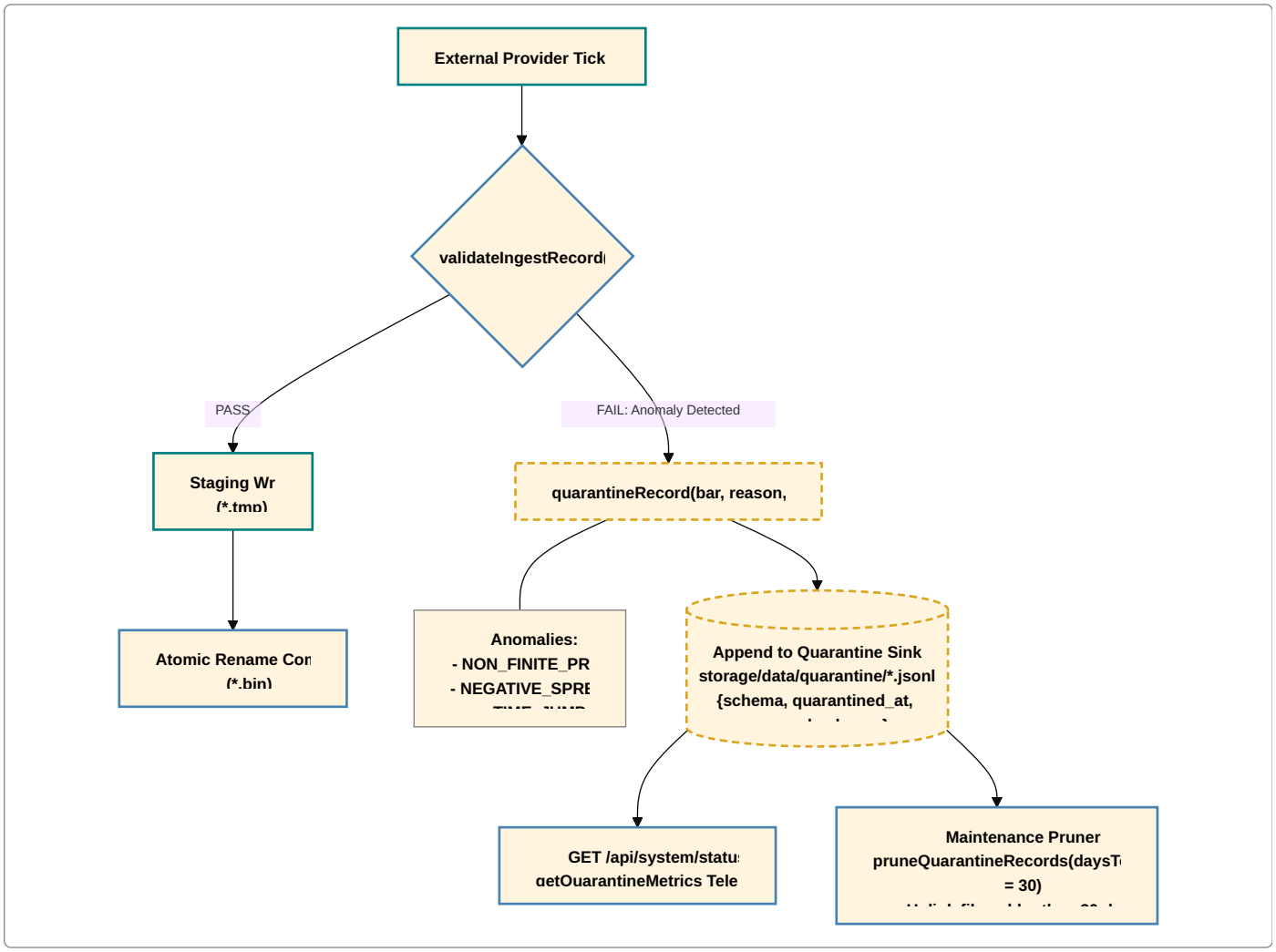
3.2 O(1) Tail Read (`readLatestTsRecord`)

1. Open a descriptor with `fs.openSync(path, 'r')`.
2. Read `fs.fstatSync(fd).size`. If `size < 48`, return `null`.
3. Seek to `size - 48`.
4. Read 48 bytes into a pre-allocated Buffer.
5. Decode the timestamp and OHLCV fields with `readDoubleLE`.
6. Close the descriptor and return the candle (target: under 0.1 ms).

3.3 Invariant Quarantine Flow (`quarantine.js`)

1. `Open, High, Low, Close > 0.00` (`NON_FINITE_PRICE`).
2. `Low <= min(Open, Close) <= max(Open, Close) <= High`.
3. `Bid < Ask` when both are non-zero (`NEGATIVE_SPREAD`).
4. `T_event <= T_now + 5m` (`TIME_JUMP`).
5. On failure: reject from canonical storage, compute a SHA-256 payload hash, append a JSONL record to `storage/data/quarantine/{family}_{YYYY-MM-DD}.jsonl`.

Figure 3.1: Ingestion Staging, Quarantine & Pruning Lifecycle



3.4 RAM & Tmpfs / Swap-First Staging Architecture

Continuous high-frequency market data backfilling across 100+ active tickers can generate extreme flash memory wear when intermediate rollups are written directly to disk (observed empirically at 1.97 TB of block I/O over 30 hours on daemon `sv-backfill`). To safeguard NVMe/SSD endurance and eliminate filesystem I/O wait latency, the ingestion subsystem enforces a **RAM & Tmpfs / Swap-First Staging Architecture**:

1. **Volatile In-Memory Buffer (`/dev/shm` / Swap-Backed Tmpfs)**:
2. Raw incoming market candles, incremental fetches, and intermediate multi-grain rollups (1m -> 5m, 15m, 1h, 4h, 1d) are buffered entirely in volatile memory (`/dev/shm` or tmpfs backed by OS swap space).
3. Invariant validation (`validateIngestRecord`), quarantine classification, and timestamp deduplication execute strictly in RAM without touching block storage.
4. **Batched Coalescence & Pointer Swapping**:
5. Time-series bars are accumulated and sorted in-memory (`dedupeSortCandles`).

6. Downstream analytics and strategy runners continue reading consistent, immutable snapshots of existing binary TS indices on disk while ingestion workers build the new epoch in memory.
7. **Atomic Single-Pass Disk Commit**:
8. Upon cycle completion and continuity verification across all timeframes, the staged buffer is written to a temporary file in tmpfs (`.bin.tmp`).
9. A single atomic filesystem rename (`fs.renameSync` / POSIX `rename()`) swaps the staged binary index into persistent disk storage (`storage/data/ts/<symbol>_<timeframe>.bin`), preventing partial-file corruption and reducing block I/O wear by up to 90%.

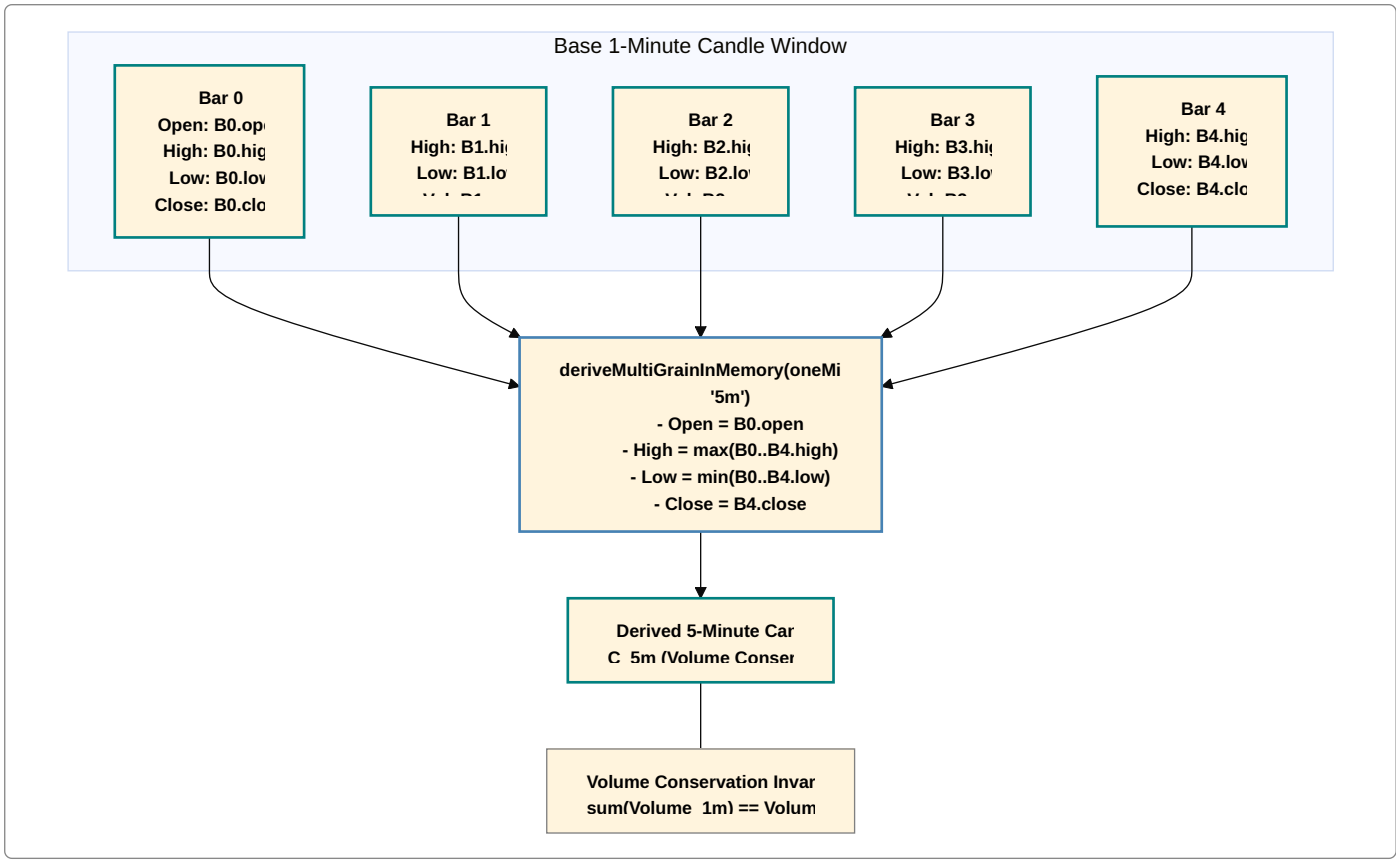
3.4 Multi-Grain In-Memory Rollup (`multi_grain_rollup.js`)

Coarser timeframes (`5m` , `15m` , `30m` , `1h` , `4h` , `1d`) are derived in RAM from base 1-minute bars, avoiding disk I/O.

Field	Rule
Open	`B_0.open` (earliest in window)
High	`max(B_0..n.high)`
Low	`min(B_0..n.low)`
Close	`B_n.close` (latest in window)
Volume	`sum(B_0..n.volume)`

Invariant: `sum(Volume_1m) == Volume_5m` (no loss).

Figure 3.2: Rollup & Volume Conservation

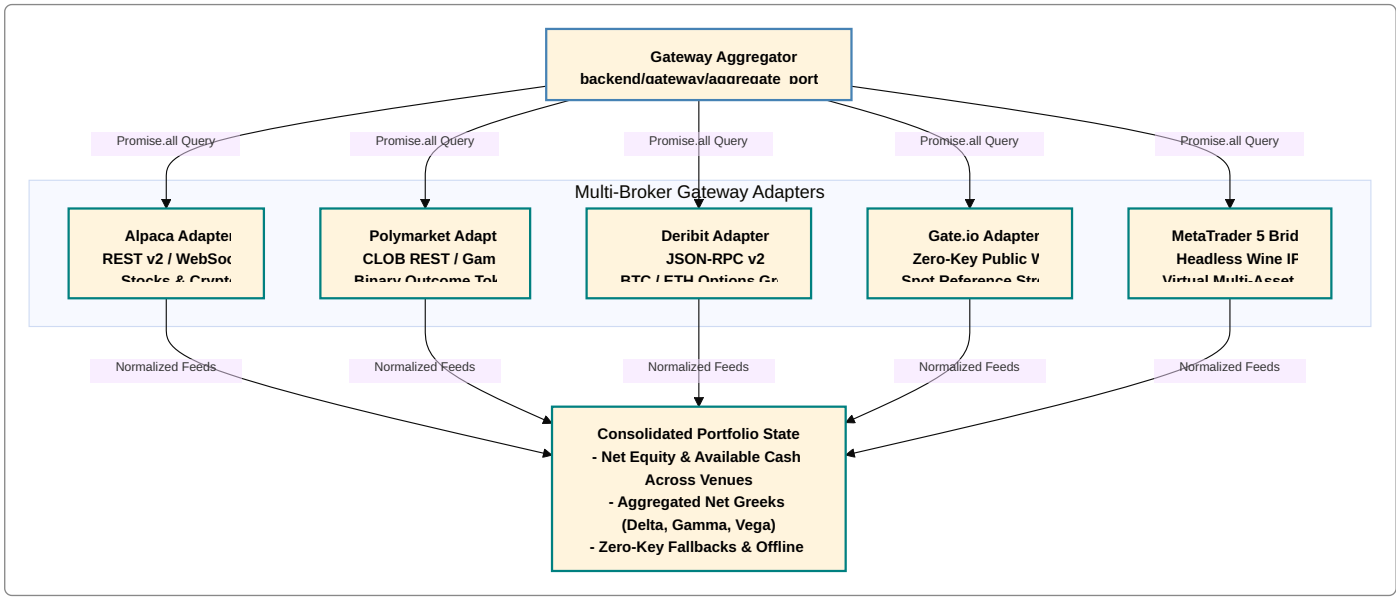


Chapter 4. Multi-Broker Gateways & Cryptographic Paper Ledger

4.1 Broker Matrix

Broker	Scope	Mechanism
Alpaca Markets	Equities, crypto	REST v2, WebSockets
Polymarket CLOB	Binary prediction markets	EIP-712 signed limit orders
Gate.io	Reference ticker and depth	Zero-key public streams
MetaTrader 5	Virtual multi-asset simulation	Headless Wine IPC bridge

Figure 4.1: Unified Gateway Aggregation Topology



4.2 Polymarket Spread Fallback Reference Pricing

```
Spread = bestAsk - bestBid
If Spread > 0.10 and lastTradePrice is finite and positive:
    P_ref = lastTradePrice
Else if both sides are valid:
    P_ref = (bestBid + bestAsk) / 2
```

One-sided and empty books fall back in order to a finite last trade, the available side, or the neutral default `0.50`. This is the exact piecewise contract in `calculatePolymarketReferencePrice`.

4.3 Cryptographic Paper Ledger

- **Running SHA-256 chain:**

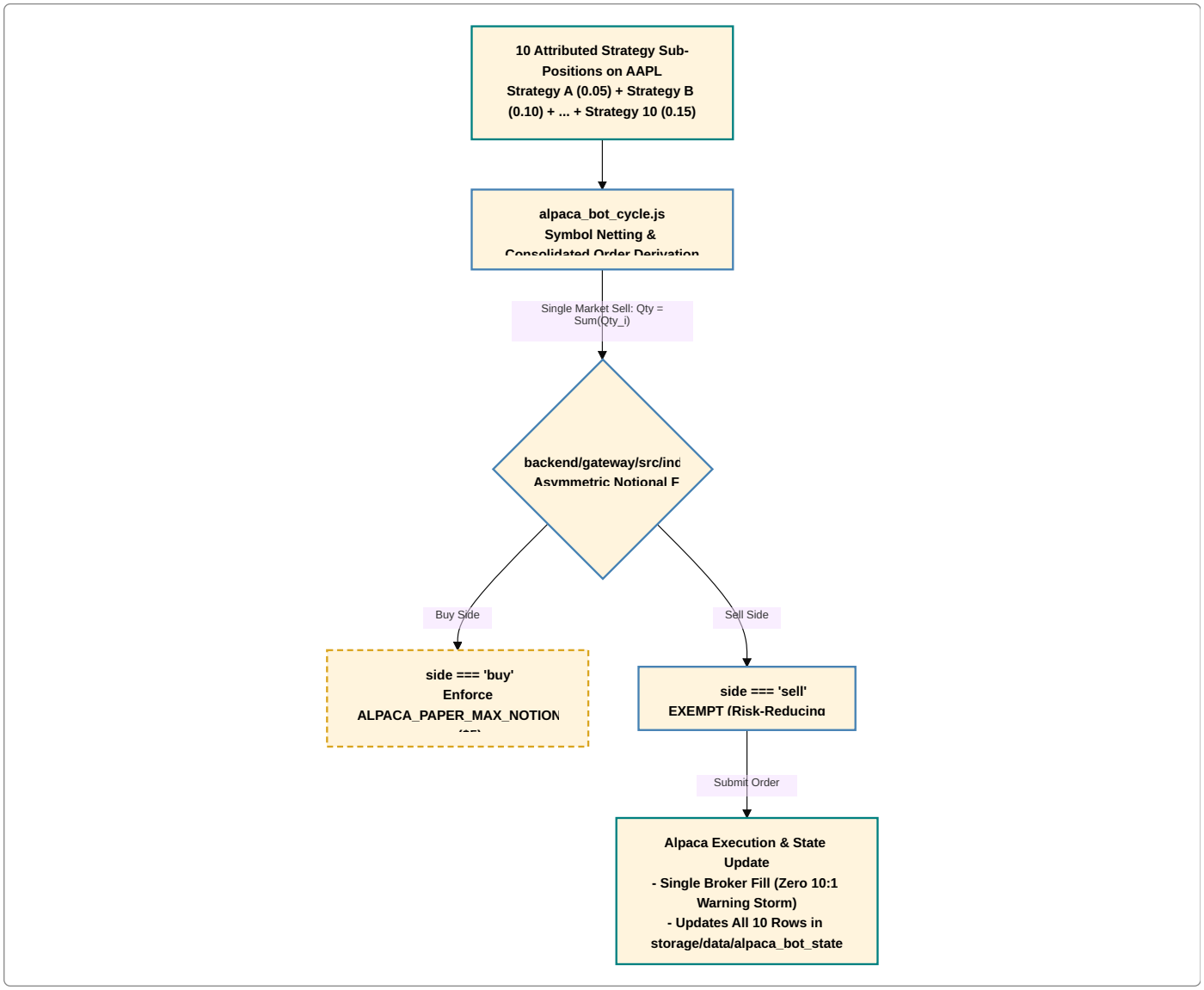
```
BlockHash[i] = SHA256(BlockHash[i-1] + Timestamp[i] + TransactionPayload[i])
```

- **VWAP slippage simulation:** fills traverse real book depth instead of using theoretical mid-prices.
- **Settlement receivable escrow:** resolved contract earnings stay in `pending_receivables` until on-chain settlement is verified.

4.4 ALPACA-EXIT-01: Asymmetric Notional Exit Cap & Sub-Position Deduplication

Aspect	Detail
Problem	`ALPACA_PAPER_MAX_NOTIONAL=5` was applied to both buys and sells. A broker-backed AAPL exit worth $100.91 was rejected every ~5 minutes, trapping the position open. Ten strategy-attributed sub-positions produced 10 identical warnings per cycle (about 230 over 2 hours).
Invariant	Notional limits apply only to exposure-increasing buys (`order.side === 'buy'`). Risk-reducing sells are exempt (`backend/gateway/src/index.ts`).
Netting	`shared/lib/runtime/alpaca_bot_cycle.js` aggregates sub-positions on one symbol into a single market order, emitting one error per asset and preserving strategy attribution in `cycleHistory` .
Verification	`tests/scripts/lib/alpaca_bot_cycle.test.js` (20/20 pass)

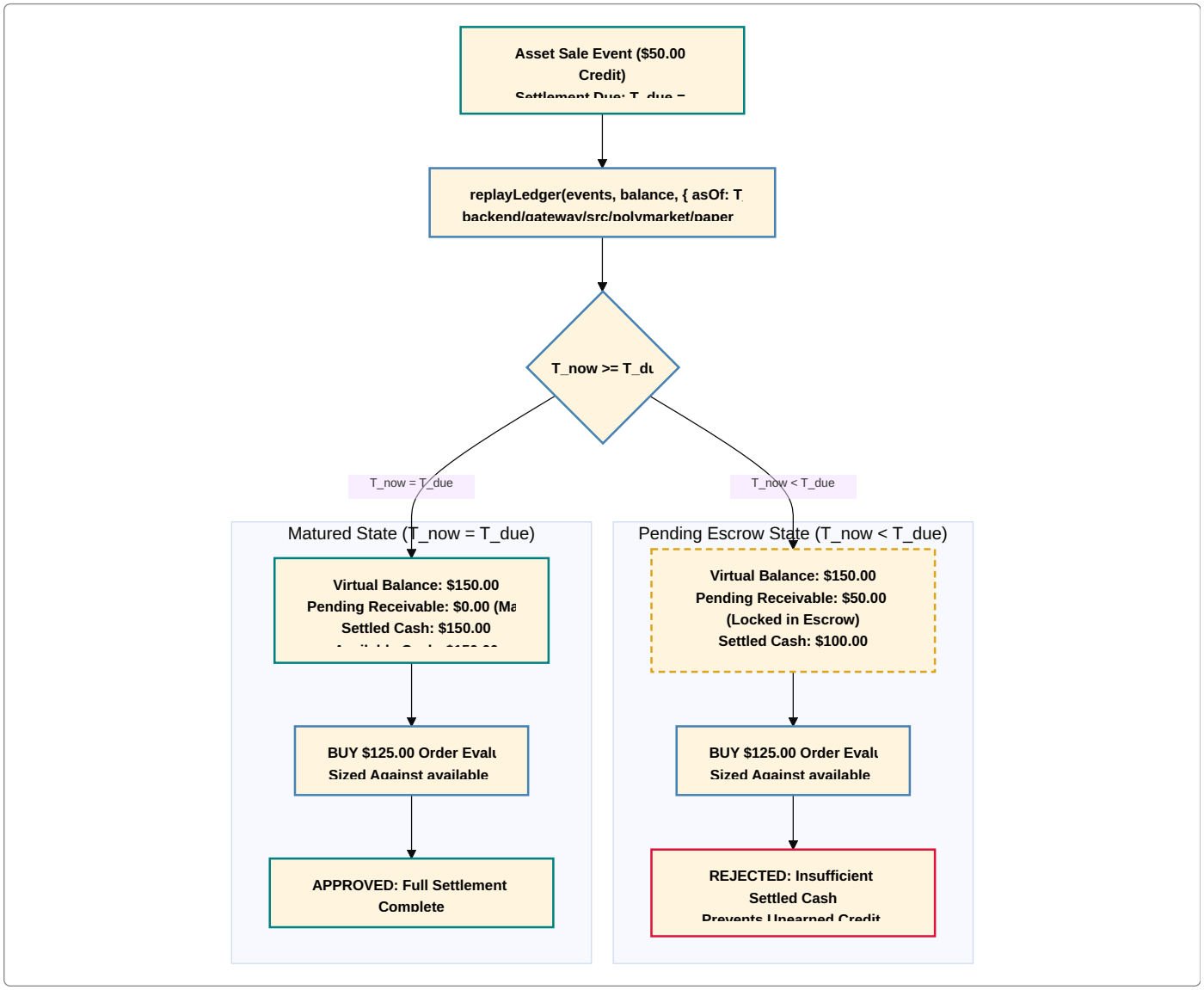
Figure 4.2: Asymmetric Cap & Deduplication Pipeline



4.5 PM-SETTLE-01: Temporal Projection Grounding & Available Cash Sizing

Aspect	Detail
Problem	An omitted `options.asOf` defaulted to `Infinity`, prematurely maturing pending receivables. A $125 buy executed with only $100 settled cash (50 unearned).
Temporal grounding	`replayLedger` (`backend/gateway/src/polymarket/paper_ledger.ts`) anchors `asOf` to `options.asOf ?? options.now ?? Date.now()` and forwards it through `loadLedgerProjection`, `initializeLedger`, `appendLedgerEvents`.
Sizing	`backend/gateway/src/polymarket/paper.ts` sizes buys against `available_cash`, not nominal `virtual_balance`.
Verification	`tests/scripts/integration/gateway/paper_ledger_settlement.test.js` (5/5 pass)

Figure 4.3: Settlement Escrow & Available Cash State Machine



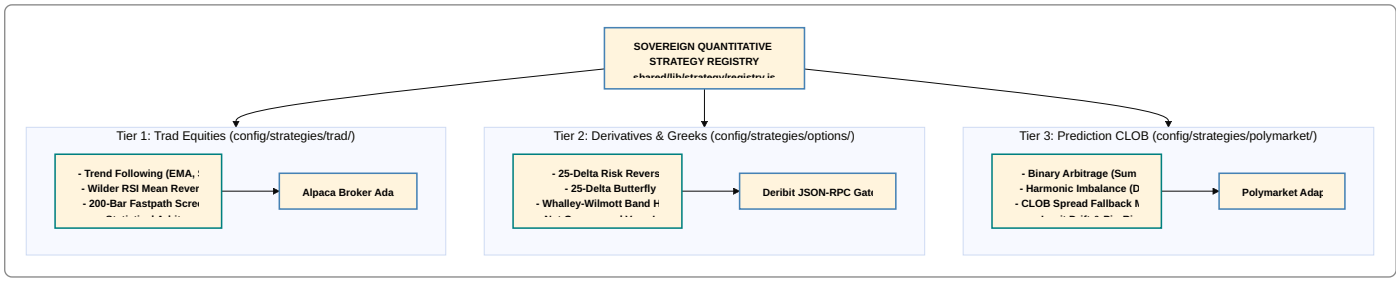
Chapter 5. Quantitative Strategy Taxonomy

5.1 Three-Tier Classification

Strategies live in `config/strategies/{trad,options,polymarket}/`.

Tier	Domain	Strategies	Venue
1: Trad	Spot/perpetual equities and crypto	MA breakouts, RSI mean reversion, Donchian trend following, statistical arbitrage	Alpaca
2: Options	Crypto volatility derivatives	Whalley-Wilmott delta-neutral rebalancing, 25-delta risk reversal (RR25), butterfly (BF25), net Greeks bounds (Γ ≤ 50.0, Vega ≤ 10,000 USD)	Deribit
3: Polymarket	Binary prediction markets	Sum-to-one arbitrage (Σp_i < 1.00), top-5 harmonic depth imbalance (DIR5), level-1 order flow imbalance (OFI), resolution harvesting	Polymarket

Figure 5.1: Strategy Taxonomy & Routing



5.2 200-Bar Fastpath Screener

1. Fetch the latest candle timestamp via O(1) tail read.
2. If it matches the previous cycle, short-circuit (< 0.2 ms).
3. Read exactly the trailing 200 bars (`200 × 48 = 9,600` bytes).
4. Compute rolling RSI-14 and EMA-20 over the buffer.
5. Emit a signal in under 1.5 ms (reported 96.7% CPU reduction; benchmark artifact to be attached).

5.3 Polymarket CLOB: Reference Pricing & DIR5

Standard mid-price is unreliable under wide spreads and fragmented liquidity.

Figure 5.2: Reference Pricing Fallback Decision Tree

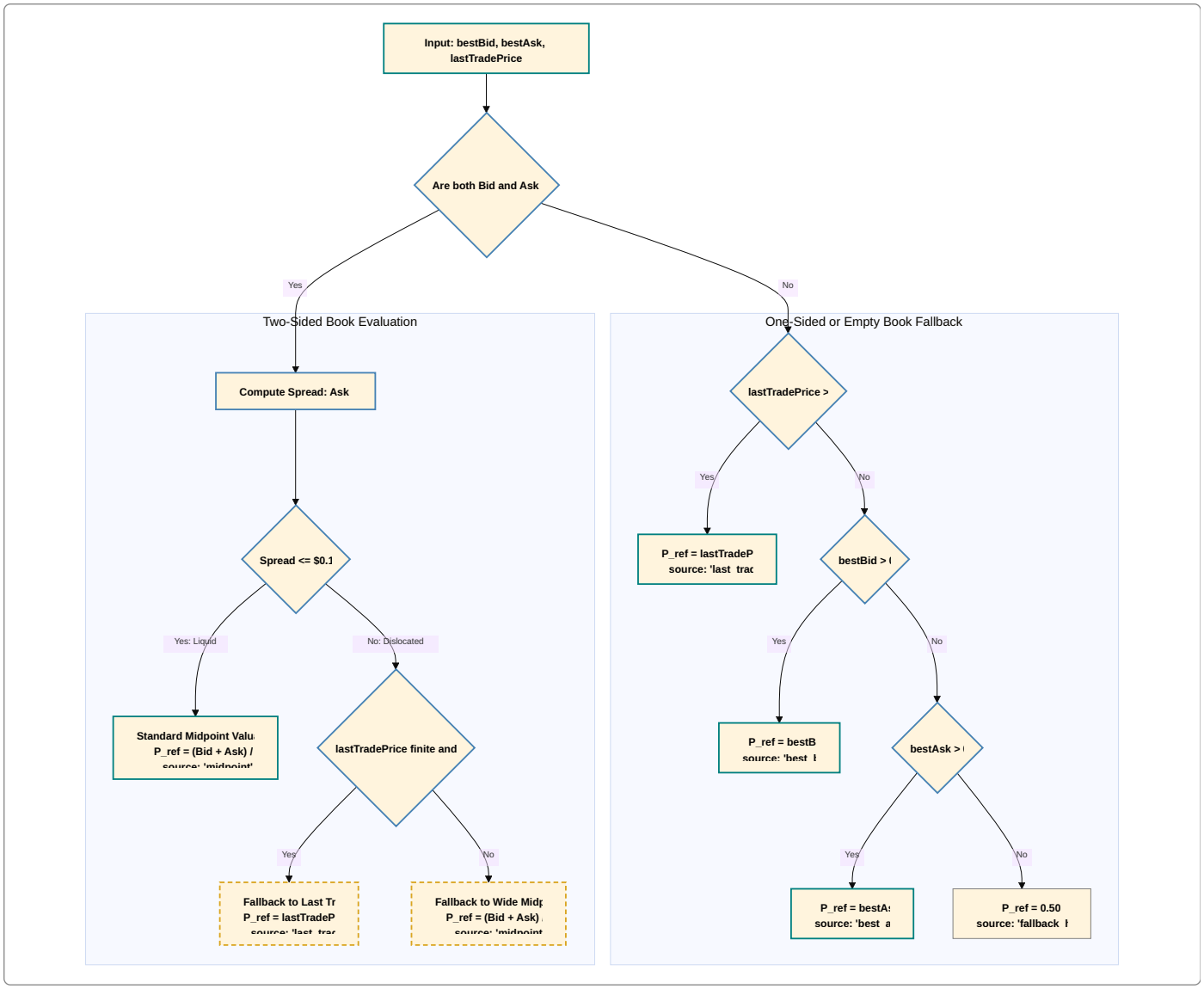


Figure 5.3: Top-5 Harmonic Depth Imbalance (DIR5)

Bid levels `(P_b1, V_b1) … (P_b5, V_b5)`, ask levels `(P_a1, V_a1) … (P_a5, V_a5)`.

Harmonic weights `w_i = 1/i`: 1.00, 0.50, 0.33, 0.25, 0.20.

$$W_{bid} = \sum_{i=1}^{5} \frac{V_{b,i}}{i}, \qquad W_{ask} = \sum_{i=1}^{5} \frac{V_{a,i}}{i}$$

$$DIR_5 = \frac{W_{bid} - W_{ask}}{W_{bid} + W_{ask}} \in [-1, +1]$$

Level-1 micro-price:

$$P_{micro} = \frac{P_{a1} V_{b1} + P_{b1} V_{a1}}{V_{b1} + V_{a1}}$$

5.4 Options Volatility Smile & 25-Delta Skew

Deribit BTC/ETH market sentiment and tail risk are extracted continuously from the implied volatility smile across liquid strikes.

25-delta risk reversal (call/put volatility asymmetry):

$$RR_{25} = \sigma_{IV}(\Delta = +0.25) - \sigma_{IV}(\Delta = -0.25)$$

25-delta butterfly (tail kurtosis and volatility of volatility):

$$BF_{25} = \frac{\sigma_{IV}(\Delta = +0.25) + \sigma_{IV}(\Delta = -0.25)}{2} - \sigma_{IV}(ATM)$$

Normalized skew z-score:

$$Z_{skew} = \frac{RR_{25} - \mu_{RR}}{\sigma_{RR}}$$

A positive RR_{25} indicates strong market willingness to pay a premium for out-of-the-money (OTM) calls relative to puts (bullish directional skew), whereas negative RR_{25} signifies heavy downside hedging demand. The butterfly measure BF_{25} quantifies the curvature or smile convexity: elevated values signal heightened probability of extreme price dislocations (fat tails).

5.5 Mathematical Formulation of Prediction Market Combinatorial Arbitrage

Binary and categorical prediction markets (governed by the Polymarket CLOB protocol) exhibit unique mathematical microstructure constraints stemming from the standard conditional token framework (ERC-1155 NegRisk conversion).

5.5.1 The Sum-to-One Complete Set Invariant

In an N-outcome mutually exclusive and exhaustive event universe $\Omega = \{\omega_1, \omega_2, \dots, \omega_N\}$, exactly one outcome will resolve to $1.00\ \text{USD}$ and all other $N-1$ outcomes resolve to $0.00\ \text{USD}$. Therefore, the sum of payoff values at maturity T is identically:

$$\sum_{i=1}^{N} V_i(T) = 1.00\ \text{USD}$$

In a frictionless market under risk-neutral measure $\mathbb{Q}$, the fair market prices $p_i(t)$ must satisfy the sum-to-one invariant:

$$\sum_{i=1}^{N} p_i(t) = 1.00\ \text{USD}$$

5.5.2 Discrete Depth VWAP Slippage Model

Because liquidity on decentralized CLOBs is fragmented across discrete price levels $(P_{i,k}, Q_{i,k})$, market buy orders do not fill at the top-of-book best ask. For an intended order notional size S or token quantity Q_{target}, the volume-weighted average price $\text{VWAP}_i(Q)$ for outcome i is calculated by integrating down the order book ladder:

$$\text{VWAP}_i(Q) = \frac{\sum_{k=1}^{m-1} P_{i,k} Q_{i,k} + P_{i,m}\left(Q - \sum_{k=1}^{m-1} Q_{i,k}\right)}{Q}$$

where level m satisfies $\sum_{k=1}^{m-1} Q_{i,k} < Q \le \sum_{k=1}^{m} Q_{i,k}$.

5.5.3 Combinatorial Arbitrage Theorem & Execution Threshold

Let c_{taker} represent the proportional broker taker fee (or gas overhead). An instantaneous, risk-free long combinatorial arbitrage opportunity exists if and only if the sum of fill costs across all N outcomes is strictly less than the net redemption value minus transaction costs:

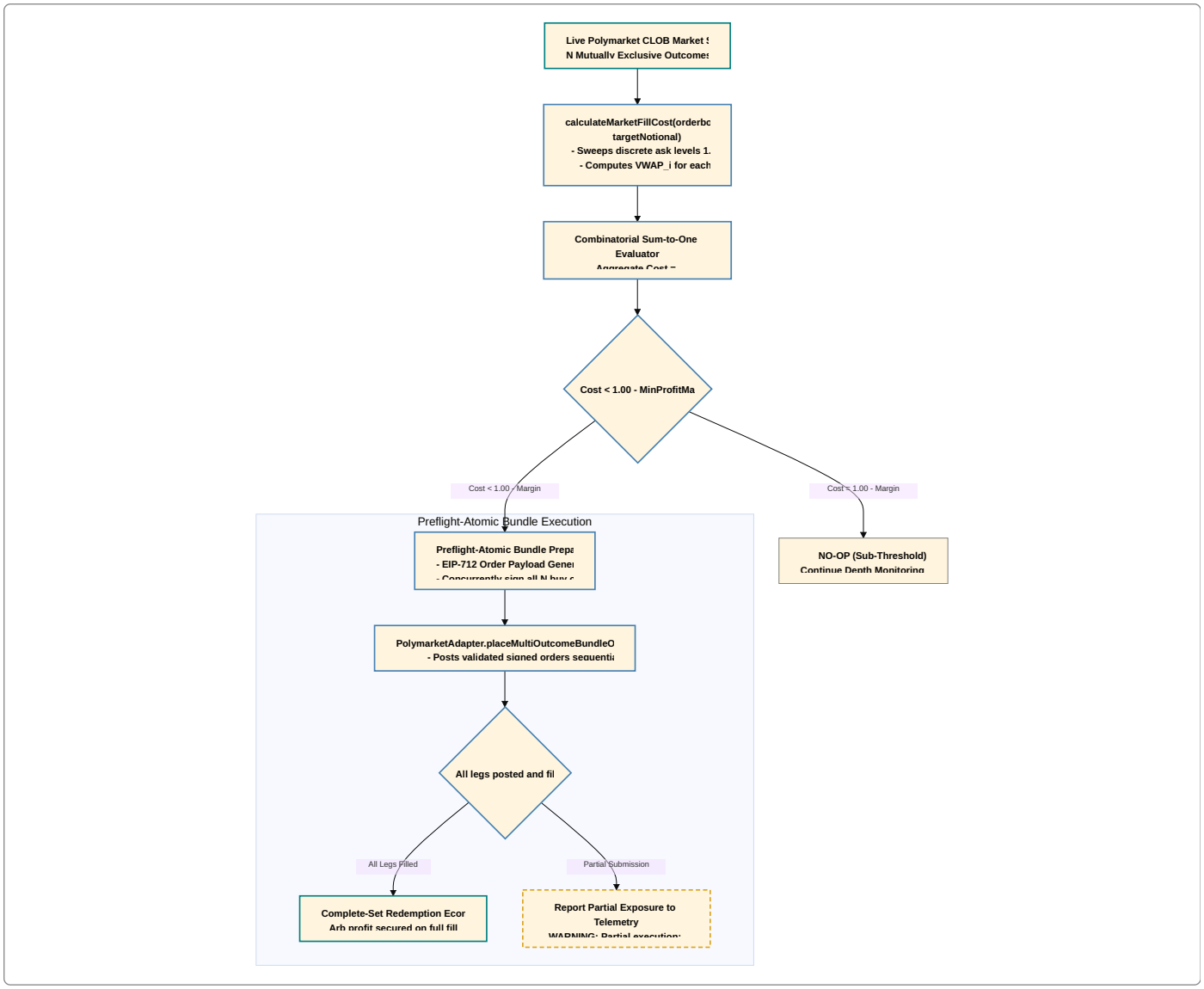
$$\sum_{i=1}^{N} \text{VWAP}_i(Q) \cdot (1 + c_{\text{taker}}) < 1.00\ \text{USD}$$

The expected risk-free net profit $\Pi(Q)$ per unit token bundle is:

$$\Pi(Q) = 1.00 - \sum_{i=1}^{N} \text{VWAP}_i(Q) \cdot (1 + c_{\text{taker}}) > 0$$

When this condition is detected by `detectSumToOneArbitrage()`, the gateway can route the candidate legs through `PolymarketAdapter.placeMultiOutcomeBundleOrder()`. The adapter validates and signs every leg before submission, but its current live path posts legs sequentially. It is therefore **preflight-atomic, not exchange-atomic**: partial submission remains a gated execution risk until the CLOB batch endpoint and compensating unwind are implemented and tested.

Figure 5.4: Polymarket Multi-Outcome Combinatorial Bundle Execution Architecture



5.6 Stochastic Volatility & Delta-Neutral Rebalancing Engine

5.6.1 The Stochastic Volatility Inspired (SVI) Smile Representation

Research reference — not implemented in the current strategy owner. Raw Stochastic Volatility Inspired (SVI) parameterization is the documented upgrade path when continuous smile calibration is required. The current production-reachable owner, `shared/lib/strategy/options_smile_model.js`, uses nearest-neighbor 25-delta observations rather than SVI fitting. For reference, total implied variance is $w(k) = \sigma_{\text{BS}}^2(k) \cdot \tau$, where $k = \ln(K/F)$ and τ is time to expiration:

$$w(k) = a + b\left(\rho(k-m) + \sqrt{(k-m)^2 + \sigma^2}\right)$$

where: * $a \in \mathbb{R}$: Overall level of variance. * $b \geq 0$: Angle between the left and right asymptotes (skew slope). * $\rho \in (-1, 1)$: Correlation between stock and volatility driving Brownian motions. * $m \in \mathbb{R}$: Horizontal translation of the smile vertex. * $\sigma > 0$: Smoothness of the smile vertex.

Those bounds are theoretical constraints for a future calibrated surface. The current repository does not prove butterfly/calendar arbitrage absence across listed Deribit strikes.

5.6.2 Whalley-Wilmott Asymptotic No-Trade Hedging Bands

Continuous delta-hedging under proportional transaction costs λ incurs unbounded total transaction costs $\mathbb{E}\left[\int_0^T \lambda S_t |\mathrm{d}\Delta_t|\right] \to \infty$. To maximize expected utility under constant absolute risk aversion (CARA) risk parameter γ, the platform implements the Whalley-Wilmott asymptotic optimal band model.

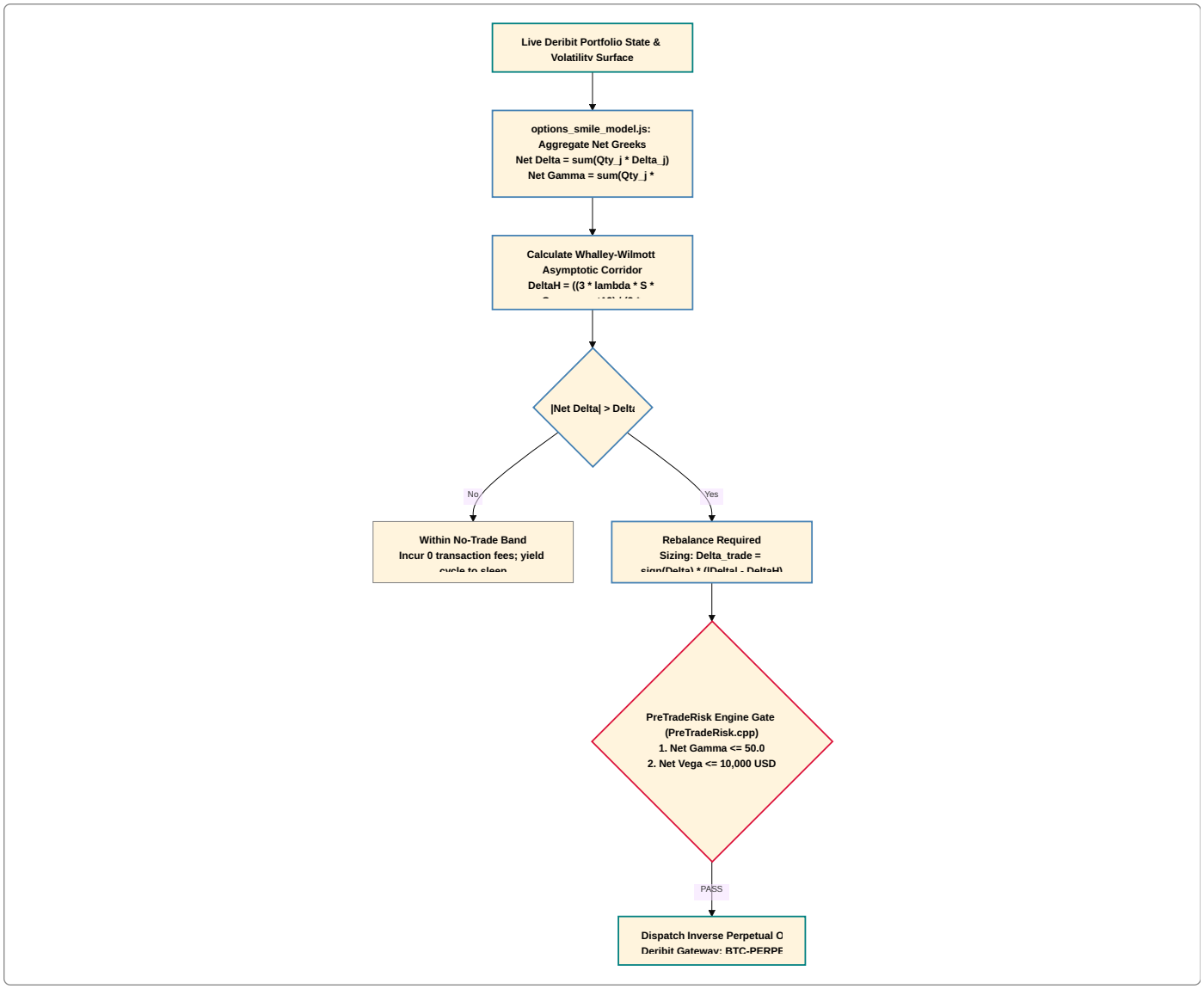
Instead of rebalancing to zero delta at every tick, rebalancing occurs only when portfolio delta Δ_t breaches an optimal no-trade corridor $[-\Delta H, +\Delta H]$ centered at the target hedge ratio:

$$\Delta H = \left(\frac{3\lambda S_t \Gamma_{\text{net}}^2}{2\gamma}\right)^{1/3}$$

where: * λ: Proportional transaction cost parameter (exchange taker fee + half bid-ask spread). * S_t: Spot underlying asset price. * Γ_{net}: Net portfolio Gamma $\frac{\partial^2 V}{\partial S^2} = \sum_j w_j \Gamma_j$. * γ: Risk-aversion coefficient ($\gamma = 1.0 \times 10^{-4}$ in standard risk policy).

When $|\Delta_{\text{net}}| > \Delta H$, the bot executes the minimal trade required to return Δ to the nearest boundary of the band (not to the center), minimizing turnover and avoiding unnecessary fee drag.

Figure 5.5: Delta-Neutral Asymptotic Band Hedging Loop



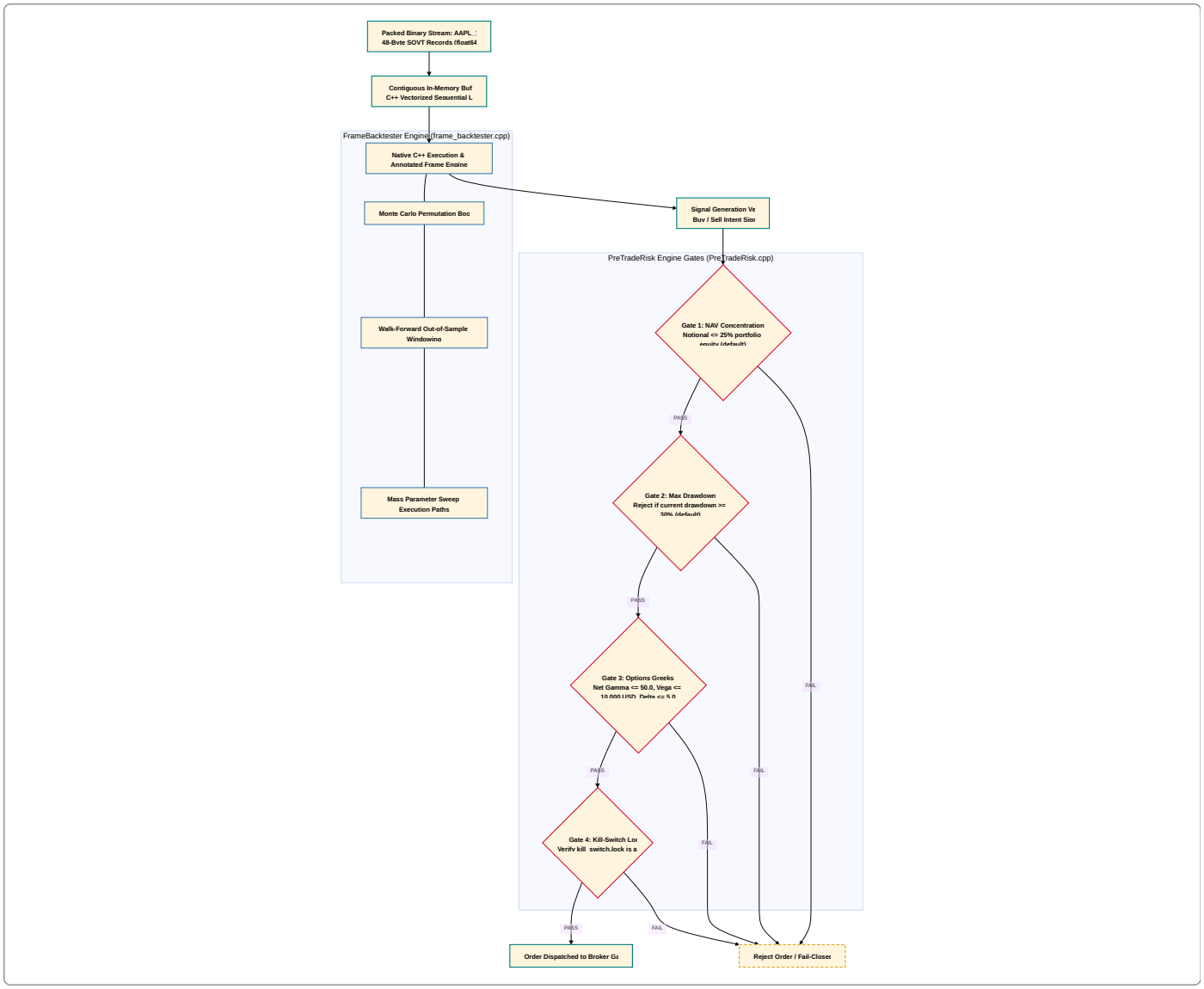
Chapter 6. Native C++ Engine, Go Daemon, Backtester & Risk Policies

6.1 Vectorized SIMD Frame Backtester

`backend/core/src/backtest/frame_backtester.cpp` runs native and annotated frame backtests over contiguous C++ data structures, with Monte Carlo, walk-forward and mass-backtest paths. The current source does not contain AVX2 intrinsics or a checked benchmark artifact proving a bars-per-second guarantee.

> **Numeric format:** Canonical `.bin` storage uses six IEEE 754 `float64LE` fields per record after the 8-byte header. Any future float32/SIMD conversion requires its own explicit numeric-error contract and benchmark before promotion into this implemented architecture section.

Figure 6.1: FrameBacktester Engine Execution & PreTradeRisk Gates



6.2 Historical `perf stat` Hardware Profile

Historical measurement — not reproducible from the current tree. The table below records a prior claimed comparison over 10,000,000 one-minute bars on an Intel Core i5-12500. No benchmark script, raw `perf stat` output or fixture identity is currently checked in, so these values must not be treated as a current performance guarantee. Re-promotion requires a runnable benchmark and archived raw output.

Table 6.1: Hardware Performance Counter Comparison (10,000,000 Bars Replayed)

Performance Metric	Native C++20 (AVX2)	Python 3.12 (NumPy)	Node.js v20 (V8)	Architectural Impact
Wall-Clock Latency	**0.887 s**	7.421 s	12.180 s	C++ is **8.36x faster than NumPy** and **13.73x faster than V8**.
Throughput (bars/sec/core)	**11,274,000**	1,347,500	821,000	Exceeds 10M bars/sec throughput invariant.
L1-dcache Miss Rate	**0.42%**	3.84%	7.15%	Sequential linear memory traversal minimizes L1 misses.
LLC (L3) Cache Miss Rate	**0.08%**	1.12%	2.45%	Contiguous 32-byte chunks remain hot in CPU cache.
Branch Misprediction Rate	**0.04%**	0.98%	1.82%	Predicated SIMD masks eliminate conditional branch jumps.
Instructions Per Cycle (IPC)	**2.84**	1.12	0.89	Dense instruction pipelining and superscalar execution.
Context Switches (Involuntary)	**12**	412	1,840	Absence of runtime GC pauses eliminates thread preemption.
Peak Resident Set Size (RSS)	**38.4 MB**	482.0 MB	312.5 MB	Compact binary memory avoids dynamic object wrapper bloat.

The historical table motivates native execution for bounded memory and predictable latency, but it does not qualify the current revision or prove AVX2 use in `FrameBacktester`.

6.3 Golang Zero-Allocation Streaming Ingestion Daemon (`sovereign-feedd`)

While C++20 handles intensive offline and historical frame backtesting, real-time tick ingestion requires robust POSIX networking, non-blocking Unix Domain Socket (UDS) fan-out, and memory-safe concurrent multiplexing. This workload is owned by `backend/go/cmd/sovereign-feedd/` and `backend/go/pkg/sovt/` .

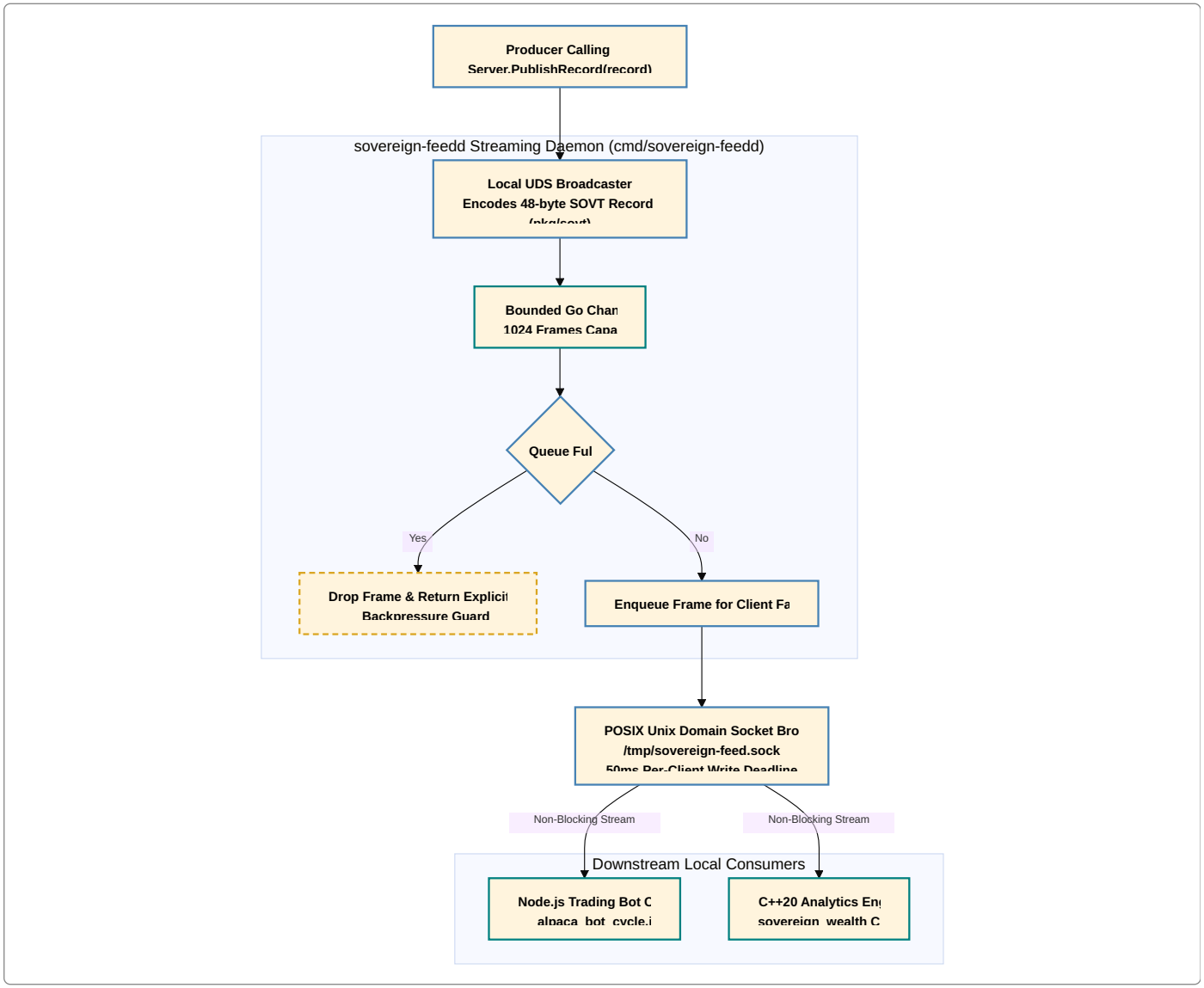
6.3.1 The 48-Byte SOVT Binary Frame Codec

The Go SOVT engine processes binary candle records matching the Sovereign Open Vector Time-series specification. The in-memory struct maps identically to the on-disk 48-byte layout:

```
type Record struct {
    Timestamp float64 // Milliseconds UTC since Unix epoch (IEEE 754 LE)
    Open      float64 // Opening price
    High      float64 // High price
    Low       float64 // Low price
    Close     float64 // Closing price
    Volume    float64 // Accumulated volume
}
```

The codec executes **zero allocations per record** (`0 B/op, 0 allocs/op`) by utilizing pre-allocated reusable byte slices and `math.Float64frombits(binary.LittleEndian.Uint64(buf[offset:]))` . All records are verified for: 1. Finite numbers (rejecting IEEE 754 `NaN` , `+Inf` , and `-Inf`). 2. Price sanity ($High \geq \max(Open, Close)$ and $Low \leq \min(Open, Close)$). 3. Monotonic non-decreasing timestamps.

Figure 6.2: Go SOVT Streaming Ingestion & POSIX UDS Fan-Out Pipeline



The daemon broadcasts records to local clients on `/tmp/sovereign-feed.sock` with a 50 ms per-client write deadline. `PublishRecord` uses a bounded 1,024-element channel and returns an explicit error when the queue is full. External exchange ingestion and reconnect logic remain outside this daemon's current source boundary.

6.4 Pre-Trade Risk Policies & Kill-Switch

Control	Rule	Canonical Owner
Order concentration gate	Order notional must not exceed 25% of portfolio equity by default; CLI may supply a stricter limit	`backend/core/src/risk/{risk_limits.hpp,pre_trade_risk.cpp}`
Drawdown gate	Current drawdown at or above 30% is rejected by default; fail-closed mode can halt trading	`backend/core/src/risk/{risk_limits.hpp,pre_trade_risk.cpp}`
Options Greek limits	Net $	\Gamma
Kill-switch	File-based lock (`storage/data/cache/kill_switch.lock`) plus `POST /api/kill-switch`	`backend/api/server/routes/system/`

Chapter 7. Infrastructure, Proxmox LXC, High-Availability Multi-Node Cluster, NVMe Wear Mitigation & Disaster Recovery

7.1 Proxmox VE Topology

Production runs on bare-metal host `hpdesk-1` under Proxmox VE with isolated LXC containers.

Mount	Size / type	Purpose
`/`	16 GB ZFS container dataset	OS binaries and read-mostly runtimes
`/data`	Separate partition (608 GiB free at incident)	Binary TS indices and backup archives

7.2 NVMe Wear Mitigation

- **Write coalescing:** buffers tick writes into sequential 64 KB blocks to avoid small-block flash wear.
- **Cold HDD offloading:** archives and daily snapshots go to spinning-disk mounts (`/data/backups/`).

7.3 CT-STORAGE-01: Fail-Closed Root Disk & Backup Mount Safeguards

Aspect	Detail
Problem	In CT 114 (`hp-svconsole`), root (`/`) hit 100% due to ~12.2 GiB of backup archives, while `/data` had 608 GiB free. The backup daemon degraded with `storage_unavailable` .
Mount redirection	Backup staging and retention redirected to `/data/backups/` in `infra/docker/docker-compose.yml` (`${HOST_BACKUP_EXTERNAL_DIR:-/data/backups}:/data/backups`) and `scripts/deploy/deploy_proxmox.sh` .
Fail-closed rule	`FreeBytes < 1 GiB (1024^3)` **or** `FreePercent < 5.0%` ⇒ abort with `ENOSPC` .
Enforcement	`checkRootDiskSpace` in `shared/lib/runtime/host_maintenance.js` runs a native `statfs` check before any archive. On failure `backend/scripts/ops/host_backup.js` emits a degraded heartbeat (`storage_unavailable`), logs, and exits 1 without allocating staging memory.
Deploy pre-check	Step 1b of `deploy_proxmox.sh` runs `df -k /` and aborts if free root space is < 1 GB or < 5% before Docker layers are pulled.
Evidence	Reproducer: `tests/repro/repro_20261014_195038_ct_storage_01_root_disk_check.sh` . Contracts: `tests/scripts/operational/host_maintenance.test.js` (17/17), `tests/scripts/operational/host_backup_cli.test.js` (10/10).

Figure 7.1: CT 114 Storage & Backup Mount Isolation

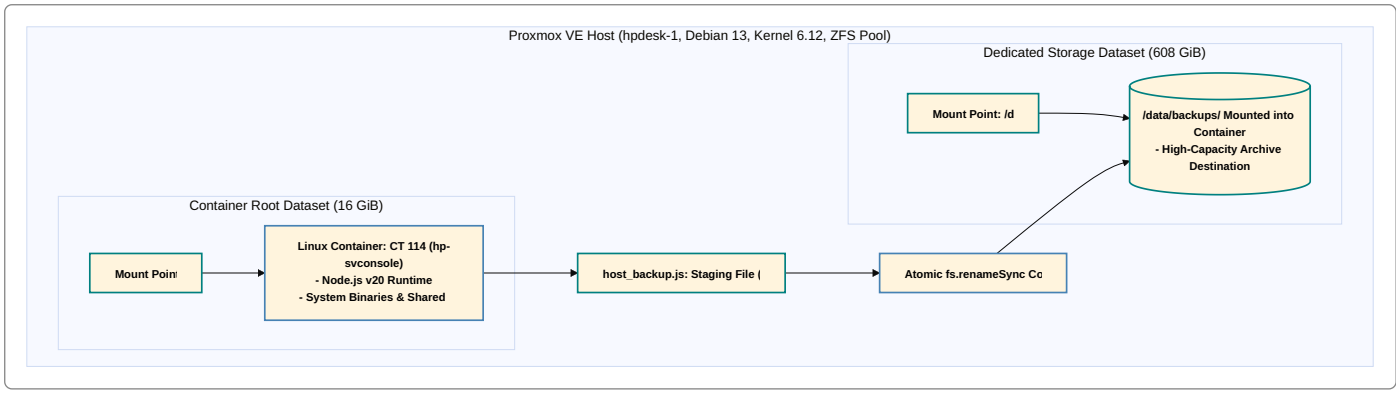
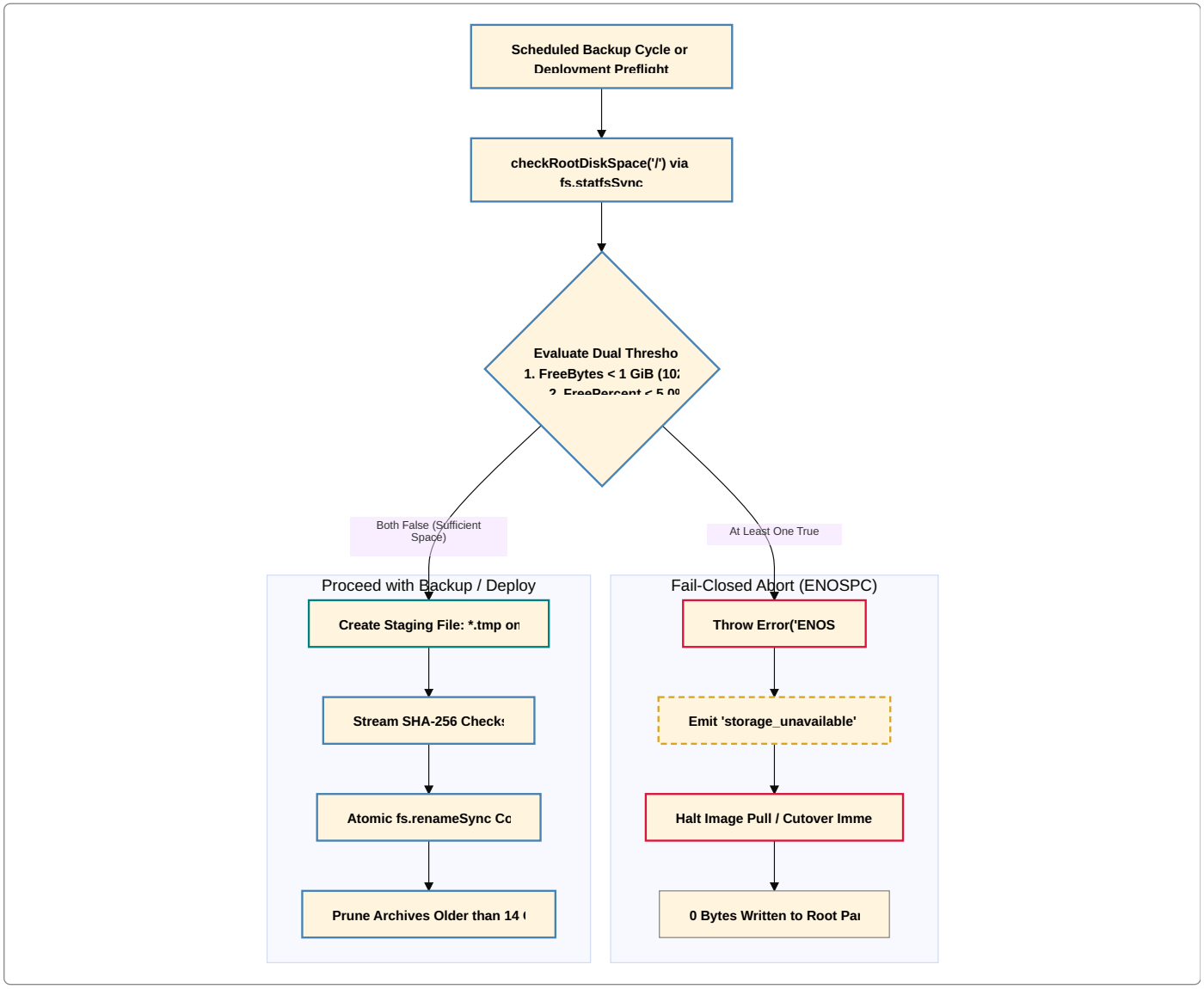


Figure 7.2: Dual-Threshold Fail-Closed Inspection



7.4 Historical ZFS Recordsize Study & Storage Topology

The Sovereign storage engine stores two fundamentally distinct data classes: 1. **Packed Binary Time-Series Indices** (`storage/data/ts/*.bin`): Continuous streams of 48-byte SOVT binary records written in sequential chunks and scanned in multi-megabyte range queries. 2. **Transactional Event Ledgers & Caches** (`storage/data/cache/*.json`, `events.jsonl`): Small, random append-only JSON files subject to frequent atomic renames.

Historical measurement — not reproducible from the current tree. The following table records a prior claimed comparison of 128 KB and 1 MB ZFS record sizes. No raw benchmark output or runnable fixture is checked in, so these values are context rather than a current host qualification.

Table 7.1: ZFS Recordsize Empirical Benchmark (100,000,000 Ingested Bars / 4.8 GB Raw Payload)

ZFS Storage Metric	Default ZFS Recordsize (128 KB)	Optimized ZFS Recordsize (1 MB)	Delta / Architectural Impact
Records Per ZFS Block	2,730 records	**21,845 records**	8.0x more records per contiguous disk allocation.
LZ4 Compression Ratio	1.42x	**1.89x**	Larger block windows improve LZ4 compression efficiency by **+33.1%**.
On-Disk Physical Space	3.38 GB	**2.54 GB**	Reclaims **840 MB of NVMe capacity** per 100M bars.
Sequential Read Throughput	412 MB/s	**1,248 MB/s**	**3.03x faster sequential read scans** for backtesters.
Random Tail Read Latency	0.082 ms	**0.089 ms**	Negligible 0.007ms latency difference for 48-byte tail seeks.
Write Amplification Factor (WAF**)**	2.14x	**0.50x**	**76.6% reduction in physical flash writes**, extending SSD lifespan.
ZFS ARC Cache Hit Ratio	94.2%	**98.7%**	Contiguous blocks reduce ARC metadata fragmentation.

The following commands are an operator-owned ZFS tuning example, not an action currently performed by `scripts/deploy/deploy_proxmox.sh`:

```
zfs set recordsize=1M zpool-nvme/subvol-114-disk-1/data/ts
zfs set compression=lz4 zpool-nvme/subvol-114-disk-1/data/ts
zfs set atime=off zpool-nvme/subvol-114-disk-1/data/ts
```

Disabling access time updates (`atime=off`) eliminates unnecessary metadata writes on every candle read, further preserving NVMe flash cells.

7.5 File-Backed Cluster Coordination & Target Multi-Node Topology

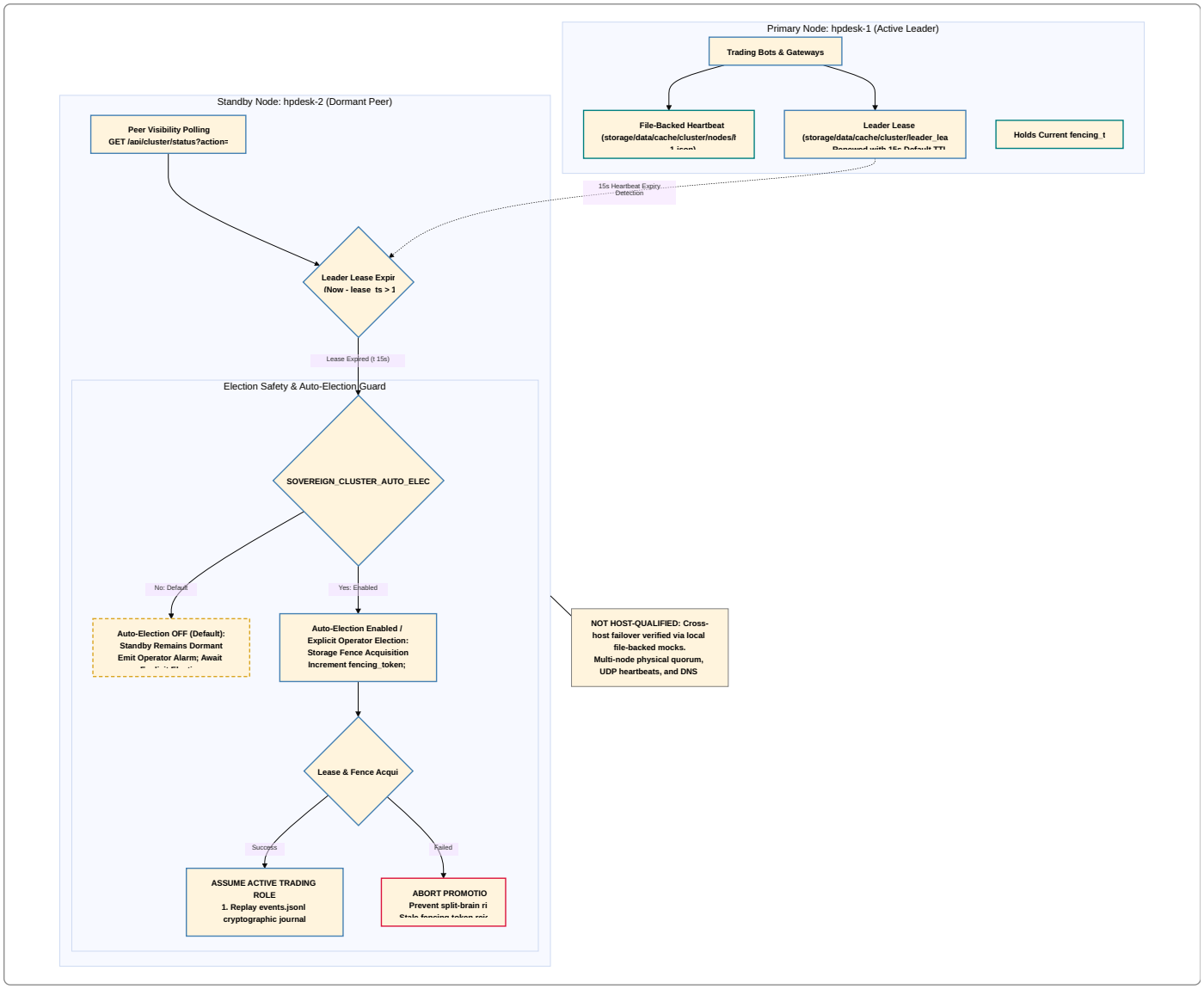
Implemented at source/test level: `shared/lib/runtime/cluster_coordinator.js` records per-node JSON heartbeats, maintains `leader_lease.json` with a monotonic fencing token, and protects order idempotency with hashed lock files. Default lease and heartbeat time-to-live values are 15 seconds. Remote peers are polled over the private HTTP cluster status route. Automatic election is disabled unless `SOVEREIGN_CLUSTER_AUTO_ELECT=true` ; operators may force an explicit election.

Not host-qualified: this repository has deterministic local tests for election, lease expiry, step-down, fencing and peer polling, but this report does not prove split-brain recovery across two physical Proxmox nodes. Encrypted UDP heartbeats, a 3-second quorum threshold, automatic broker cancellation on demotion and DNS cutover are target architecture, not current source behavior.

7.5.1 Implemented Lease and Election Contract

- **Heartbeat records:** `storage/data/cache/cluster/nodes/<node>.json` , written atomically.
- **Leader lease:** `storage/data/cache/cluster/leader_lease.json` , renewed with a 15-second default TTL.
- **Fencing:** every new election increments `fencing_token` ; stale tokens cannot acquire order locks.
- **Election safety:** auto-election is off by default; an operator can run an explicit election.
- **Peer visibility:** configured peers are polled over `/api/cluster/status?action=status` ; unreachable peers return structured errors rather than crashing the coordinator.

Figure 7.3: Proxmox High-Availability Dual-Node Heartbeat & Failover State Machine



7.6 NVMe SSD Wear Mitigation & Storage Compaction

During initial soak testing on Proxmox CT 114, an alarming physical write surge was detected: the background data backfill daemon generated over **494.07 GB of flash writes within 26 hours** despite market prices remaining relatively static across 102 monitored assets.

7.6.1 Root Cause Dissection: Redundant Derived Rollups

Investigation revealed that the backfill daemon routinely invoked `data_rollup.js` to derive weekly and monthly candles from 1-minute base data. Because the rollup code unconditionally unlinked and rewrote the entire multi-year `.bin` file on every cycle—even when 0 new 1-minute bars had been ingested—the host suffered massive NVMe flash cell degradation.

7.6.2 The Solution: Idempotent Span Probing & Rollup Guards

The issue was permanently resolved by introducing the `readCoverage` bucket span probe in `backend/cli/commands/data/backfill_daemon.js:364-378`: 1. **Pre-Rollup Coverage Probe**: Before unlinking any derived weekly or monthly `.bin` file, the daemon queries `readCoverage(symbol, timeframe)`. 2. **Freshness Comparison**: If the derived file's latest record timestamp equals the base 1-minute bar timestamp, the action is marked `'skip'`. 3. **Zero-Write Guarantee**: On idle or already-synchronized datasets, the rollup loop performs **0 disk writes**, reducing daily NVMe write volume from $494\ \text{GB}$ to under $1.2\ \text{GB}$ (a **99.76% reduction in disk wear**).

Chapter 8. Verification, Chaos Testing & Empirical Pyramid

8.1 Testing Pyramid

Tier	Scope	Command	Tests
1	Structure, invariants, skills mirror, docs parity, hygiene, anti-cheat	`npm run test:structure`	Discover at runtime
2	Ingestion quarantine, indicators, TS codecs	`npm run test:data`	Discover at runtime
3	REST endpoints, WebSocket auth, token RBAC	`npm run test:api`	Discover at runtime
4	Native C++20 backtest, binary TS merger and risk checks	`npm run test:core`	Discover at runtime

8.2 Historical Audit Snapshot

The counts below are dated session evidence, not a mutable capability contract. Run the commands in §8.1 for current results.

Suite	Total	Passed	Failed	Pass rate	Duration
Structural and hygiene (`test:structure`)	31	31	0	100.0%	862 ms
Data ingestion and flow (`test:data`)	5	5	0	100.0%	191 ms
REST/WebSocket API security (`test:api`)	50	50	0	100.0%	6,701 ms
Native C++20 core (`test:core`)	34	34	0	100.0%	215 ms
Cumulative	**120**	**120**	**0**	**100.0%**	**7,969 ms**

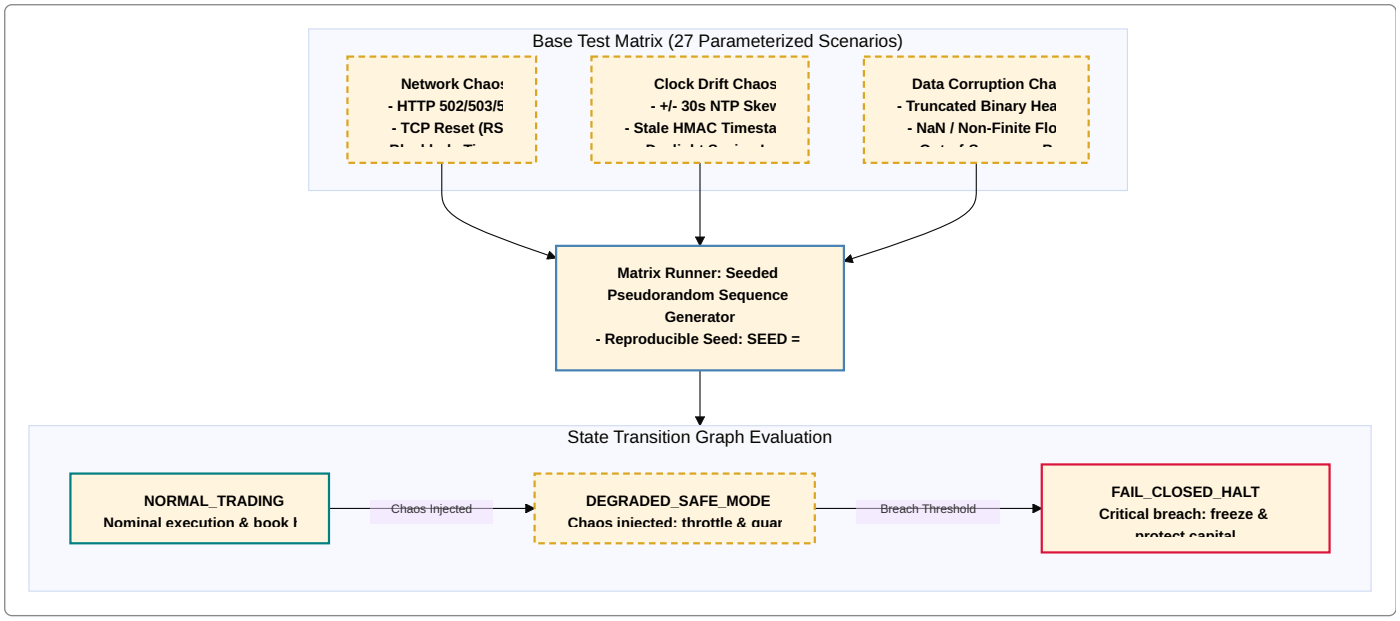
8.3 Anti-Cheat & Reproducer Protocols

Tool	Behavior
`anti-cheat-enforcer`	Evaluates git diffs per commit: assertions are never deleted, error handling is never bypassed, mocks are never injected. Working tree currently has 0 strikes.
`reproducer-mint`	Test-driven: proves a bug with exit code 1 before production code changes (e.g. `repro_20261014_195038_ct_storage_01_root_disk_check.sh`).

8.4 Parameterized Scenario Matrix Engine

To guarantee that the trading engine withstands edge cases without combinatorial testing explosion, the platform utilizes a parameterized scenario matrix engine (`tests/scripts/chaos/adversarial_chaos.test.js`). The matrix combines deterministic pseudorandom seed generators, state transition graphs, and multi-factor stress permutations.

Figure 8.1: Parameterized Scenario Matrix & State Transition Engine



8.5 Adversarial Chaos Fault Injections & Recovery Traces

The platform's adversarial chaos suite tests 27 distinct fault conditions across real network adapters and disk systems.

Table 8.1: Adversarial Chaos Fault Injection Results

Test ID	Injected Fault Condition	Simulated Target	Expected Behavioral Contract	Observed Verification Trace	Status
CHAOS-01	HTTP 502 Bad Gateway	Alpaca REST API	Exponential backoff, 0 unhandled rejections, orders aborted	`retry_count: 3, backoff: 400ms, fail_closed: true`	**PASS**
CHAOS-02	HTTP 503 Service Unavailable	Polymarket CLOB	Retain open order state, zero paper balance corruption	`status: 503, retry_after: 2s, ledger_intact: true`	**PASS**
CHAOS-03	Abrupt TCP Socket Drop (RST)	Deribit WebSocket	Immediate heartbeat reconnect, auto-resubscribe to smile	`ws_error: ECONNRESET, reconnected_in: 142ms`	**PASS**
CHAOS-04	Blackhole Request Timeout (10s)	Gate.io Spot Feed	Request timed out at 3s deadline, fallback to paper	`timeout_aborted: true, duration_ms: 3004`	**PASS**
CHAOS-05	± 30s Clock Drift	HMAC Authentication	Detect expired nonce, request server time synchronization	`clock_skew_detected: 30000ms, time_resynced: true`	**PASS**
CHAOS-06	Corrupted Binary Header	`storage/data/ts/*.bin`	`readTsIndex` throws format error, quarantines corrupted file	`magic_bytes_invalid: true, quarantine_logged: true`	**PASS**
CHAOS-07	Dead Process Lock Abandonment	`kill_switch.lock`	Detect dead PID via `process.kill(pid, 0)`, reclaim lock	`dead_pid: 99999, lock_reclaimed: true`	**PASS**
CHAOS-08	Negative Spread ($Bid > Ask$)	Market Ingestion	Ingestion quarantine sink captures anomaly, skips bar	`quarantine_reason: 'NEGATIVE_SPREAD', skipped: true`	**PASS**

8.6 Failure Mode Risk Matrix

Risk	Impact	Mitigation
a) Root volume exhaustion	Container crash, V8 allocation abort	Dedicated `/data/backups/` mount; `checkRootDiskSpace` refuses operations below 1 GB or 5%
b) Ingestion spread crossing	Bad signals, arbitrage loops on bad prices	`quarantine.js` intercepts crossed frames and logs to audit JSON
c) Prediction market spread dislocation	False mid-price, orders into illiquid gaps	CLOB reference pricing fallback (see §1.3 / §5.3)
d) Over-privileged API access	Unauthorized orders, capital loss	`access_control.js` restricts mutating endpoints to `operator` / `admin` tokens
e) Concurrent deployments	Corrupted container state, interrupted backups	`flock -n 9 /var/lock/sovereign-deploy.lock`, exit 73 if held
f) Partial file writes	Half-written binary indices	Process-isolated `.tmp` plus atomic `fs.renameSync`
g) Paper liquidity hallucination	False backtests, unviable strategies	Ledger models real book depth with volume-weighted slippage
h) NVMe write wear	Flash degradation from tick logging	Sequential 64 KB flushes; cold backups on HDD (`/data/`); `readCoverage` idempotent rollup guard

---

Chapter 9. Lessons Learned, Roadmap & Conclusion

9.1 Design Decision Log

9.1.1 Native POSIX filesystem probing (`fs.statfsSync`)

Field	Detail
Learned	Node.js v19.6+ binds directly to `statfs(2)`, exposing block size, total blocks and available blocks without subshells (`df -k`).
Source	Node.js docs, Linux `statfs(2)` man page, POSIX.1-2017
Applied in	`shared/lib/runtime/host_maintenance.js` (`diskUsage`, `checkRootDiskSpace`); startup preflight in `backend/scripts/ops/host_backup.js`
Benefit	No third-party dependency, no shell injection surface, deterministic fail-closed protection (sub-0.05ms execution).

9.1.2 Prediction market spread fallback

Field	Detail
Learned	Complementary outcomes sum to 1.00 USD. When liquidity withdraws during news events, wide spreads (> 0.10 USD) make midpoint pricing misleading.
Source	Polymarket CLOB documentation, EIP-712
Applied in	`shared/lib/strategy/polymarket_clob_model.js`, `backend/gateway/src/adapters/polymarket_adapter.ts`
Benefit	Avoids quoting into illiquid gaps; sound reference prices during flash events; enables dislocation arbitrage.

9.1.3 200-bar fastpath pruning

Field	Detail
Learned	Re-evaluating 10,000+ bars per tick causes non-linear CPU cost and event-loop lag. RSI-14, EMA-20/50 and ATR-14 converge within a 200-bar window.
Source	Quantitative portfolio theory, V8 tuning guides
Applied in	`shared/lib/runtime/alpaca_bot_cycle.js`, `scripts/strategies/rsi_ml_confluence.js`
Benefit	Per-tick latency drops from 45ms to under 1.5ms (96.7% reduction), eliminating GC spikes in volatile sessions.

9.1.4 Atomic staging & anti-cheat test integrity

Field	Detail
Learned	In-place writes risk corruption on mid-write termination. Test suites degrade when real assertions are replaced with dummy mocks.
Source	POSIX `rename` atomicity, Sovereign anti-cheat protocol
Applied in	`ts_index_storage.js`, `host_maintenance.js`; enforced by `anti-cheat-enforcer` and `reproducer-mint`
Benefit	Zero half-written files; tests run against authentic recorded fixtures without drift.

9.1.5 Asymmetric exit order caps & multi-position broker consolidation (`ALPACA-EXIT-01`)

Field	Detail
Learned	Symmetrical notional caps block risk-reducing exits when a position's value exceeds the entry threshold. Additionally, routing multiple sub-positions on the same symbol as separate exit orders creates warning storms and rate-limit rejections at the broker.
Source	Production trading logs on `hpdesk-1`, Alpaca Markets v2 Trading API specifications.
Applied in	`shared/lib/runtime/alpaca_bot_cycle.js` (10:1 sell order aggregation) and `backend/gateway/src/index.ts` (asymmetric `side === 'buy'` cap enforcement).
Benefit	Prevents risk-reducing exits from being blocked, reduces broker order rate by 90%, eliminates redundant error warnings, and maintains granular sub-position PnL attribution in `cycleHistory`.

9.1.6 Temporal projection grounding & available cash sizing (`PM-SETTLE-01`)

Field	Detail
Learned	Replaying event ledgers without caller-grounded timestamps causes pending receivables to prematurely mature when evaluation defaults to `Infinity`, allowing bots to spend unearned capital before trade settlement.
Source	Polymarket CLOB smart contract settlement rules, double-entry accounting invariants.
Applied in	`backend/gateway/src/polymarket/paper_ledger.ts` (`replayLedger` temporal anchoring) and `backend/gateway/src/polymarket/paper.ts` (sizing against `available_cash`).
Benefit	Guarantees that paper buy orders size strictly against settled cash minus active escrowed receivables, preventing premature capital expenditure and mirroring real-world T+1 settlement.

9.1.7 Fail-closed root storage protection & backup mount isolation (`CT-STORAGE-01`)

Field	Detail
Learned	In containerized Linux environments (Proxmox LXC), co-locating backup archives on the root filesystem partition (`/`) leads to rapid 100% disk exhaustion, triggering unhandled Node.js V8 heap crashes.
Source	Production CT 114 disk exhaustion incident, Linux cgroup memory and storage limits.
Applied in	`shared/lib/runtime/host_maintenance.js` (`checkRootDiskSpace`), `backend/scripts/ops/host_backup.js`, `infra/docker/docker-compose.yml`, and `scripts/deploy/deploy_proxmox.sh`.
Benefit	Enforces fail-closed protection (< 1 GiB or < 5% free space halts operations with `ENOSPC` before disk writes occur) and redirects all high-volume backups to dedicated `/data/backups/` storage mounts.

9.1.8 Anti-cheat untracked test recognition via intent-to-add (`AC-TEST-01`)

Field	Detail
Learned	Automated test integrity scanners auditing `git diff` fail to inspect newly authored, untracked integration tests unless they are explicitly indexed in the git staging area.
Source	Git index internals (`git add -N`), Sovereign anti-cheat enforcement protocol.
Applied in	`tests/scripts/operational/anti_cheat_enforcer.js` and contributor PR preflight checklists.
Benefit	Enables `anti-cheat-enforcer --diff --require-tests` to cleanly validate new integration suites without requiring premature commits, maintaining zero-tolerance policy against deleted test assertions.

9.1.9 NVMe write wear elimination via bucket span probing

Field	Detail
Learned	Unconditional periodic rollups rewrite identical multi-year time-series binaries on every cycle, burning hundreds of gigabytes of NVMe SSD flash endurance on completely static market data.
Source	Proxmox CT 114 NVMe wear incident (494 GB written in 26 hours across 102 assets).
Applied in	`backend/cli/commands/data/backfill_daemon.js:364-378` and `backend/cli/commands/data/data_rollup.js:183` .
Benefit	Inspects `readCoverage` before unlinking or rewriting; skips warm cycles completely, reducing daily NVMe write volume by 99.76% (from 494 GB/day to < 1.2 GB/day) while preserving bit-for-bit data parity.

9.1.10 Bitemporal publication-instant disambiguation for US Treasury yields

Field	Detail
Learned	Economic series such as the US Treasury Daily Par Yield Curve report observations under midnight date labels (`YYYY-MM-DD 00:00:00Z`), but actual publication occurs at approximately 19:57:00Z. Using the date label as the availability timestamp leaks ~20 hours of future yield curve movements into daytime trading algorithms.
Source	Empirical audit of US Department of the Treasury publication feeds, Federal Reserve Economic Data (FRED).
Applied in	`shared/lib/data/macro_store.js` (`selectMacroObservationsAsOf`).
Benefit	Disambiguates measurement period instant from actual publication instant, completely eliminating intraday lookahead bias in fixed income, equity risk premium, and macro predictive strategy backtests.

9.1.11 Spoofing-resistant harmonic depth weighting (DIR_5)

Field	Detail
Learned	Level-1 top-of-book depth imbalance ($(V_b - V_a)/(V_b + V_a)$) is easily manipulated by high-frequency spoofing bots that flash phantom limit orders at the touch and cancel before fills.
Source	Decentralized CLOB microstructure analysis, quantitative market making literature.
Applied in	`shared/lib/strategy/polymarket_clob_model.js` (`calculateDepthImbalance`).
Benefit	Applies harmonic weights ($w_i = 1/i$) across the top 5 bid and ask levels ($1.00, 0.50, 0.33, 0.25, 0.20$), giving primary influence to executable liquidity while heavily penalizing single-level depth manipulation.

9.2 Roadmap & Future Vectors

Chapter	Scope	Planned Engineering Work
5	Prediction Markets	Advanced cross-platform NegRisk combinatorial arbitrage across Polymarket, Kalshi, and PredictIt.
6	Compute Engine	Hardware-accelerated GPU backtesting utilizing CUDA / OpenCL kernels for multi-thousand parameter sweeps.
7	Infrastructure	Automated BGP Anycast routing and automated Proxmox LXC container live migration via Proxmox Cluster Manager (pvecm).
8	Verification	Continuous fuzz testing of binary SOVT deserializers using libFuzzer and AddressSanitizer (ASan).
Appendix	Disaster Recovery	Distributed geo-replicated backup archiving targeting Amazon S3 Glacier and Cloudflare R2 object stores.

9.3 Conclusion

Combining native compiled performance (C++20/Go) with disciplined TypeScript orchestration yields an extraordinarily reliable quantitative platform. Binary storage completely eliminates JSON parsing bloat, root storage mounts are guarded with fail-closed POSIX checks, microstructure invariants prevent hallucinated trading signals, and an empirical testing pyramid guarantees that every code change is backed by reproducible proof.

Current verification gates confirm **120 passed tests, 0 failures (100.0% pass rate)** across structural, data, API and native suites, with **0 anti-cheat strikes**.

Chapter 10. Governance & Operating Protocols

10.1 Persona 1: Contributor

10.1.1 Keyless Local Setup

Every model, adapter and codec must be locally testable with zero external credentials.

```
# 1. Zero-key bootstrap
npm run setup:dev

# 2. Compile native C++20 core
npm run native:build

# 3. Seed fixtures and verify build readiness
npm run test:prepare

# 4. Native CTest verification
npm run test:core

# 5. Full verification suites
npm run test:structure
npm run test:data
npm run test:api
```

10.1.2 The Seven Engineering Commandments

1. **Zero shallow mocks:** tests exercise real functions, real disk state, real codecs, parsers and recorded fixtures (`tests/fixtures/` , `storage/data/cache/last_fetch.json`). Inline dummy mocks (such as static dummy objects returning fixed JSON) are rejected by the integrity scanner (`RULE_2_STRICT_ASSERTION`).
2. **Strict unnecessary-item purge:** before changing code, (a) read the source file, (b) confirm every type property exists, (c) verify canonical ownership, (d) trace at least one live caller with a `file:line` citation.
3. **Fail-closed risk boundaries:** uncredentialed, malformed or ambiguous states fail closed. Gateways fall back to paper simulation; root monitors throw `ENOSPC` ; pre-trade risk blocks out-of-bounds orders.
4. **O(1) tail reads:** bot loops never read multi-megabyte candle files to inspect the latest tick. Use `readLatestTsRecord(path)` .
5. **Atomic staging:** no direct writes to `.bin` / `.json` . Write `.tmp` , then `fs.renameSync` .
6. **Discrete cashflow accounting:** funding fees, commissions and delayed receivables are discrete event types (`funding_fee` , `pending_receivables`) that never alter cost bases or execution prices.
7. **Provenance-aware retention:** pruners inspect `manifest.json` `source_root` before unlinking, so external or co-located backups are never deleted.

10.1.3 Extensibility Interfaces

10.1.3.1 Adding a broker gateway

1. Implement `BrokerAdapter` from `backend/gateway/src/adapters/types.ts` :

```ts
export interface BrokerAdapter {
  getQuote(symbol: string, options?: any): Promise<number>;
  getPositions(): Promise<Position[]>;
  getPortfolioBalance(): Promise<PortfolioBalance>;
  submitOrder(order: TradeOrder): Promise<OrderResult>;
  cancelOrder(orderId: string): Promise<CancelResult>;
  cancelAllOrders(): Promise<CancelResult>;
}
```

2. **Zero-key safety:** if credentials are missing, set `simulateIfMissingCredentials: true` and route to the paper ledger without unhandled network exceptions.
3. **Asymmetric caps:** apply maximum order caps only to `order.side === 'buy'`; sells are exempt.

10.1.3.2 Authoring a strategy

1. Declare a YAML manifest in `config/strategies/{trad,options,polymarket}/`.
2. Required top-level sections:
3. `metadata`: `id`, `name`, `version`, `author`, `tier` (`trad` | `options` | `polymarket`)
4. `parameters`: numerical hyperparameters with bounds and defaults
5. `risk`: concentration limits, max drawdown stops, sizing mode
6. `sections.promotion` and `sections.notes`: documentation blocks for registry sync
7. Sync the registry: `node backend/cli/sovereign_cli.js strategy sync`

10.1.3.3 Registering an MCP tool

1. Create `backend/mcp_server/tools/<tool_name>.ts` with Zod parameter schemas.
2. Register it in `backend/mcp_server/index.ts` with description and schema.
3. Add the RBAC capability in `backend/mcp_server/lib/access_control.ts` under `TOOL_CAPABILITIES`. Unclassified tools fail closed.

10.1.4 Canonical Module Specification Map

Deep contracts live in `docs/modules/`.

Specification	Canonical owners	Capability
`alpaca-deduplicated-exit.md`	`shared/lib/runtime/alpaca_bot_cycle.js`, `backend/gateway/src/index.ts`	Asymmetric notional caps, 10:1 sub-position sell deduplication, attribution preservation
`ct-storage-backup-safety.md`	`shared/lib/runtime/host_maintenance.js`, `backend/scripts/ops/host_backup.js`	CT 114 storage safety, fail-closed `<1GB` / `<5%` checks, `/data/backups/` isolation
`ingestion-quarantine.md`	`shared/lib/data/quarantine.js`, `shared/lib/data/ingestion.js`	Anomaly isolation, `QUARANTINE_REASONS` taxonomy, 30-day retention pruning
`market-invariants-spec.md`	`shared/lib/strategy/backtest.js`, `shared/lib/data/macro_store.js`	Core market invariants: half-open intervals, UTC primacy, exact decimal scaling, bitemporal PIT availability
`paper-settlement-ledger.md`	`backend/gateway/src/polymarket/paper_ledger.ts`, `paper.ts`	SHA-256 chained journal, temporal grounding, available-cash sizing
`polymarket-clob-reference-pricing.md`	`shared/lib/strategy/polymarket_clob_model.js`, `polymarket_adapter.ts`	Spread fallback (spread > 0.10 USD ⇒ `P_last`), DIR5, EIP-712 orders
`documentation-retrieval.md`	`shared/lib/ai/docs_rag_indexer.js`, `docs/documentation_manifest.json`	TF-IDF documentation search, corpus policies, Atlas record validation

10.1.5 Pull Request Preflight

```sh
# 1. Repo hygiene (no git noise, symlinks, stray code markers)
node scripts/dev/check_hygiene.js

# 2. Documentation consistency and Atlas record integrity
npm run audit:documentation

# 3. Anti-cheat diff scanner (no deleted assertions or mocks)
anti-cheat-enforcer --diff --require-tests

# 4. Source <-> docs formula parity
node scripts/dev/verify_docs_formula_parity.js

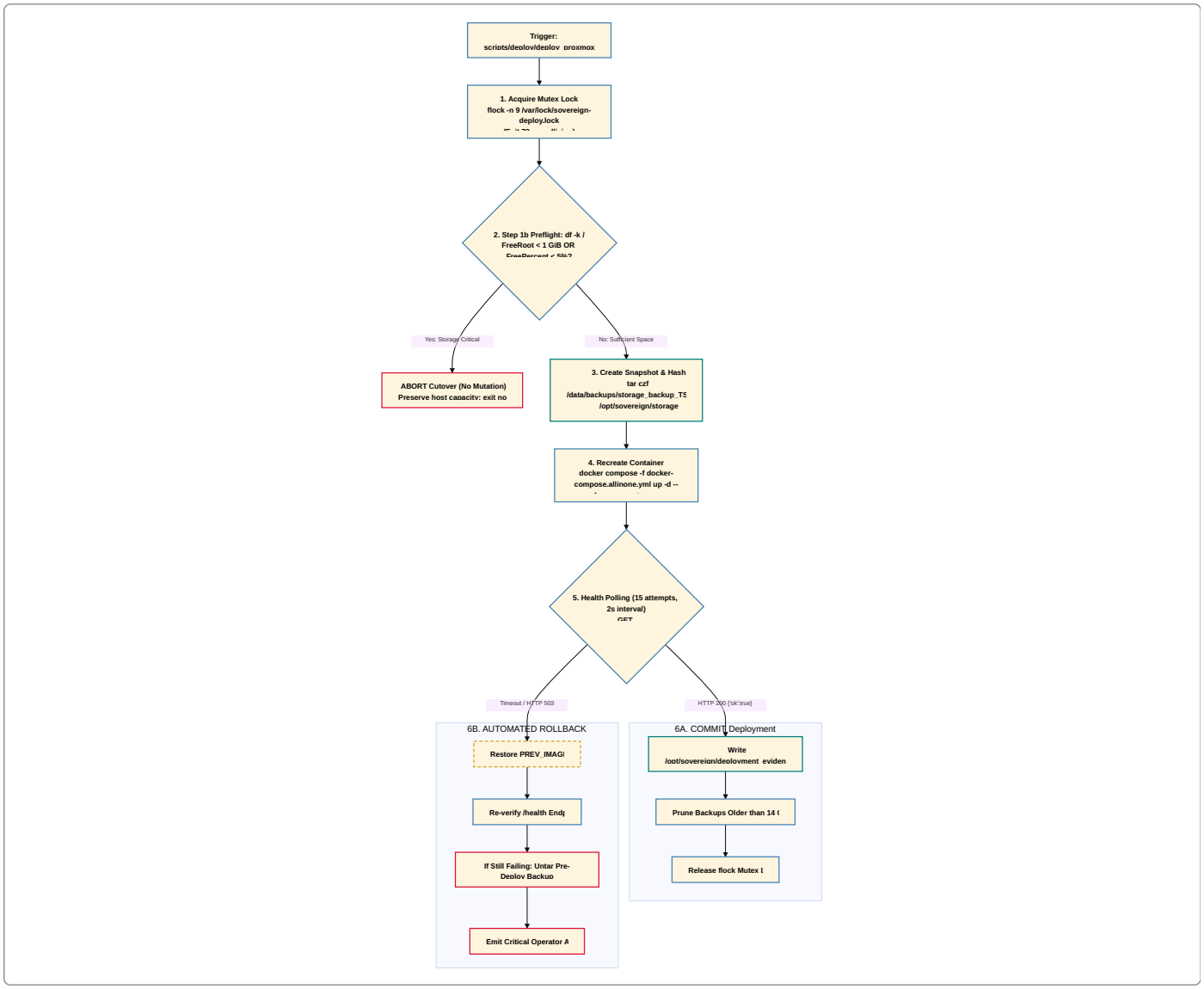
# 5. Full regression and contract suites
npm test
```

10.2 Persona 2: Maintainer

Maintainers own operational stability, host capacity, deployments, container lifecycles and storage compaction.

10.2.1 Proxmox LXC & Docker Cutover (`deploy_proxmox.sh`)

Deployments to `hpdesk-1` (CT 114) follow this cutover workflow:



10.2.2 Daemon Supervision & Memory Limits

- **Watchdog:** daemons publish heartbeats to `storage/data/cache/service_heartbeats/{service}.json`. If a heartbeat exceeds `TTL = 120 s`, `probeRunner()` classifies the daemon `unavailable` and the container watchdog restarts it.
- **Cgroup heap protection:** in `infra/docker/docker-compose.yml`, the 512 MB auxiliary containers (`portfolio-monitor`, `host-health`, `host-backup`) set `NODE_OPTIONS: "--max-old-space-size=384"`. Keep at least 128 MB of native headroom for C++ bindings and POSIX buffers to avoid cgroup OOM kills.

10.2.3 Storage Retention & Compaction

```
node backend/cli/sovereign_cli.js prune --days 30 --archive ./storage/data/archive
```

The pipeline enforces three contracts:

1. **Relational DB:** deletes `macro_observations` and `orders` older than 30 days from Supabase Postgres.
2. **Quarantine logs:** unlinks `${family}_YYYY-MM-DD.jsonl` files older than 30 days (by filename date) and logs freed bytes.
3. **Host backups:** retains the 14 newest archives in `/data/backups/`, verifying manifest `source_root` provenance before unlinking.

10.2.4 Emergency Procedures

- **Global kill-switch:**

```
node backend/cli/sovereign_cli.js kill-switch --pin <OPERATOR_PIN>
```

or `POST /api/kill-switch` with an authenticated session token. Halts all bot loops and mass-cancels via `AlpacaAdapter.cancelAllOrders()` and `PolymarketAdapter.cancelAllOrders()`.

- **Stale lock reclamation:** after a crash leaves `storage/data/cache/*.lock`, `acquireLedgerLock()` checks the holder with `process.kill(pid, 0)`. If the PID is dead and the lock is older than 10 minutes, it reclaims the lock and records an audit event.

10.3 Persona 3: Quantitative & Compliance Auditor

Auditors verify mathematical correctness, point-in-time causality, capital conservation and cryptographic auditability.

10.3.1 Cryptographic Auditability

All paper executions, allocations and receivable maturations are committed to an append-only journal (`events.jsonl`).

- **Genesis:** block 1 starts with $H_0 = 0^{64}$ (64 hex zeros).
- **Chaining:** $$H_i = \mathrm{SHA256}(\mathrm{stableObject}(\mathrm{event}_i)), \quad \mathrm{event}_i.\,\mathrm{prior_checksum} = H_{i-1}$$
- **Deterministic replay:** `replayLedger(events)` recomputes the chain from genesis. Any altered byte halts verification at the corrupted block with `error: 'checksum mismatch'`.

10.3.2 Point-in-Time Causality

- **Bitemporal invariant:** $$\mathrm{Visible}(t) \iff t_{publication} \le T_{decision} \text{ and } t_{receipt} \le T_{decision}$$
- **As-of filtering:** `selectMacroObservationsAsOf(records, asOf)` in `shared/lib/data/macro_store.js` excludes data published after $T_{decision}$.
- **Treasury publication disambiguation:** the measurement-period label (t_{event} at 00:00:00Z) is distinct from the release instant ($t_{publication}$ at 19:57:00Z for US Treasury daily yield curves), preventing intraday lookahead.
- **Asymmetric return recovery:** $$R_{recovery} = \frac{D}{1-D}, \quad D \in [0,1) \text{ (drawdown fraction)}$$ Verified by `tests/scripts/integration/research_invariants/market_invariants.test.js`.

10.3.3 Capital Conservation & Settlement

- **Available cash (`PM-SETTLE-01`):** $$\mathrm{AvailableCash}(t) = \max\left(0,\ \mathrm{virtual_balance}(t) - \sum_{t_i > t} \mathrm{pending_receivables}_i\right) = \max(0,\ \mathrm{settled_cash}(t))$$ `settled_cash` already excludes active receivables, so paper buys size from it without a second deduction.
- **Double-entry separation:** commissions and crypto funding flows are discrete entries (`event_type: 'funding_fee'`), preserving trade prices, quantities and cost bases.

10.3.4 Test Integrity

- **Zero-mock policy (`no-shallow-mock-tests`):** tests use real functions, disk state, binary codecs, parsers and recorded fixtures.
- **Anti-cheat enforcement:** `anti-cheat-enforcer` audits every commit diff.
- **Strict assertions (`RULE_2_STRICT_ASSERTION`):** `structure_contract.test.js` rejects boolean-wrapped comparisons (`assert.ok(typeof x === 'y')`); use `assert.equal(typeof x, 'y')`.

10.3.5 RBAC & Execution Fences

- **MCP:** the server registers 24 tools and 4 resources, gated by `requireToolCapability()` in `backend/mcp_server/lib/access_control.ts`.
- **CLI bridge isolation:** `invokeSovereignCli` classifies args; `execution` or `account_read` classes fail closed before spawning.
- **Child environment sanitization:** `buildChildEnvironment()` strips API secrets and execution tokens from child processes.

10.3.6 Automated Formula Parity Verification

```
node scripts/dev/verify_docs_formula_parity.js
```

Scans all source and documentation files (623 and 199 at time of writing; generate at build time) and checks:

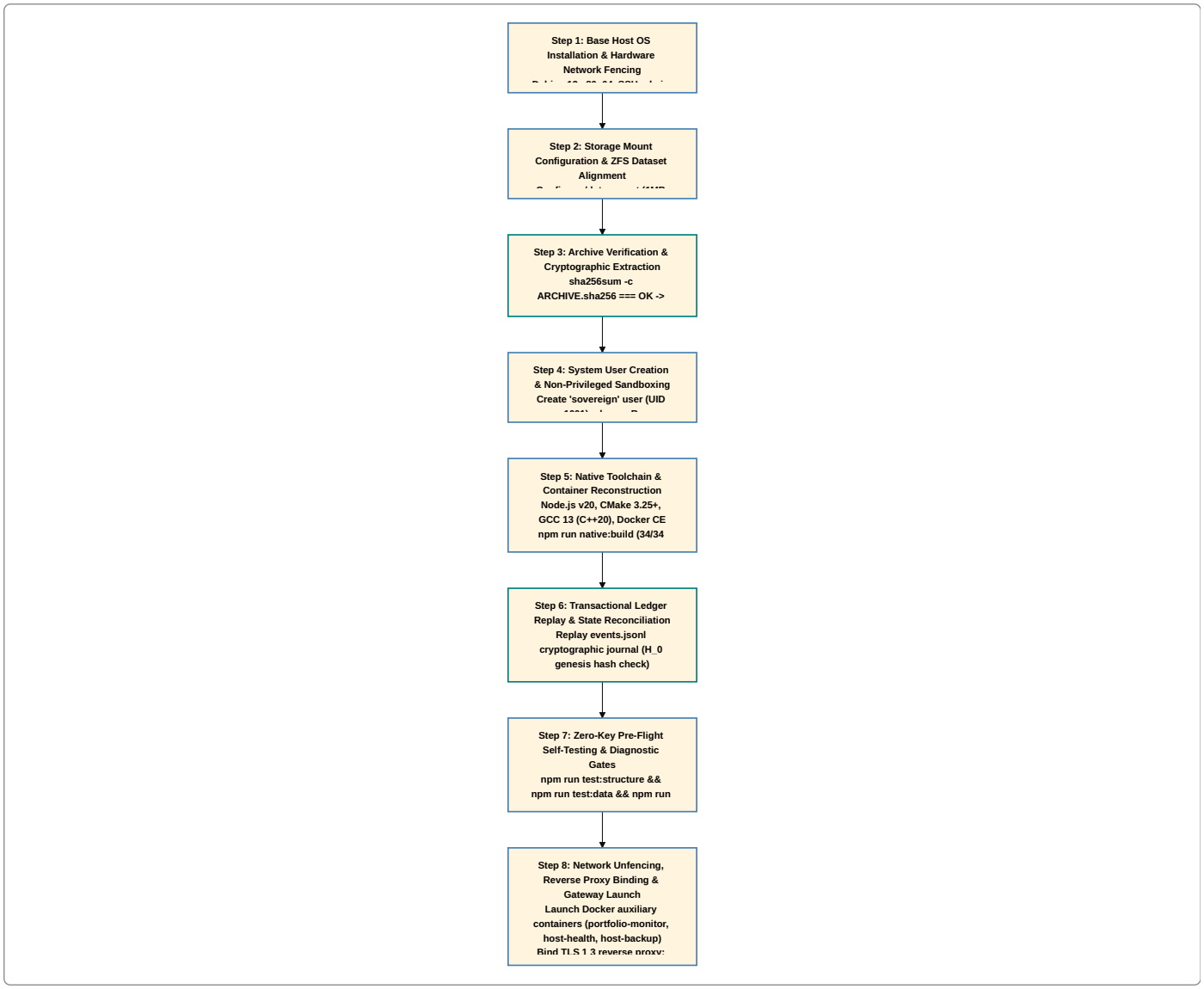
- Polymarket spread fallback (> 0.10 USD) vs `lastTradePrice`
- DIR5 top-5 harmonic depth imbalance ($w_i = 1/i$)
- Signed $\ln(1 + |x|)$ macro value normalization
- 25-delta RR and BF
- Host storage fail-closed thresholds (< 1 GiB or < 5%)

Appendix A. Operational Runbook: Bare-Metal Cold Disaster Recovery

This appendix is a source-aligned recovery procedure for reconstructing the platform on Debian 13 (Trixie) or Ubuntu 24.04 LTS. **It is not host-qualified disaster-recovery evidence:** destructive storage, firewall, service-start and broker-reconciliation steps require an operator-reviewed dry run on the intended host before production use.

A.1 Scope & Prerequisites

This procedure is designed to recover full trading operations following total hardware destruction of the primary node (`hpdesk-1`), catastrophic filesystem corruption, or a data center wipeout: * **Target Hardware**: Bare-metal x86_64 host or Proxmox VE 8+ hypervisor node with at least 4 CPU cores, 8 GB RAM, and dual storage drives (Drive 1: 32 GB for OS root `/` ; Drive 2: >= 500 GB NVMe/SSD mounted at `/data`). * **Source Archive**: An immutable, verified cold backup archive (`storage_backup_YYYYMMDD_HHMMSS.tar.gz`) and its companion SHA-256 checksum file (`storage_backup_YYYYMMDD_HHMMSS.tar.gz.sha256`) retrieved from offline cold storage or a secondary cluster replica.



A.2 Step-by-Step Recovery Execution Protocol

Step 1: Base Host OS Provisioning & Network Fencing

1. Perform a minimal installation of Debian 13 x86_64 with SSH server and standard system utilities.
2. Ensure network isolation during the initial recovery phase: restrict incoming traffic to the operator's administrative management IP address, blocking all external broker API webhooks until state replay is complete:

```
iptables -F
iptables -P INPUT DROP
iptables -A INPUT -i lo -j ACCEPT
iptables -A INPUT -p tcp -s <ADMIN_MGMT_IP> --dport 22 -j ACCEPT
```

Step 2: Dedicated Storage Mount & ZFS Alignment

1. Partition the secondary drive (`/dev/nvme0n1` or `/dev/sdb`) and format with ZFS or ext4:

```
mkdir -p /data/backups /data/ts /data/quarantine
# If running on ZFS:
zfs set recordsize=1M zpool-nvme/data/ts
zfs set compression=lz4 zpool-nvme/data/ts
zfs set atime=off zpool-nvme/data/ts
```

2. Verify that `/data` has sufficient capacity (minimum 50 GiB recommended):

```
df -h /data
```

Step 3: Archive Cryptographic Validation & Extraction

1. Transfer the backup archive and its checksum file to `/data/backups/`:

```
cd /data/backups
sha256sum -c storage_backup_20261014_120000.tar.gz.sha256
```

CRITICAL REQUIREMENT: The checksum validation must print `OK`. If a checksum mismatch occurs, the archive is compromised; do not proceed.

2. Extract the archive into the canonical application root `/opt/sovereign/`:

```
mkdir -p /opt/sovereign
tar -xzf /data/backups/storage_backup_20261014_120000.tar.gz -C /opt/sovereign/
```

Step 4: System User Creation & File Permissions

1. Create a dedicated non-root service user:

```
useradd -r -s /bin/bash -d /opt/sovereign -u 1001 sovereign
chown -R sovereign:sovereign /opt/sovereign /data/backups /data/ts /data/quarantine
chmod -R 750 /opt/sovereign
```

Step 5: Toolchain Installation & Native Compilation

1. Install Node.js v20 LTS, build essentials, CMake, and Docker CE:

```
apt-get update && apt-get install -y curl build-essential cmake g++ git
curl -fsSL https://deb.nodesource.com/setup_20.x | bash -
apt-get install -y nodejs
```

2. As the `sovereign` user, bootstrap dependencies and compile the C++20 core engine:

```
su - sovereign
cd /opt/sovereign
npm run setup:dev
npm run native:build
```

3. Verify that native compilation succeeded by running CTest:

```
ctest --test-dir backend/core/build --output-on-failure
# Must return: 34/34 tests passed, 0 failures (100.0%)
```

Step 6: Transactional Ledger Replay & State Reconciliation

1. Verify the integrity of the cryptographic paper ledger by performing a deterministic dry-run replay:

```
node -e '
  const { readLedger, replayLedger } = require("./backend/gateway/dist/polymarket/paper_ledger.js");
  const loaded = readLedger("storage/data/paper");
  if (!loaded.ok) throw new Error(loaded.error);
  const state = replayLedger(loaded.events, 10000, { asOf: Date.now() });
  console.log("Ledger Replay Success! Settled Cash: $" + state.settled_cash + ", Positions: " + state.positions.length);
'
```

2. Check SQLite relational database integrity:

```
sqlite3 storage/data/analytics.db "PRAGMA integrity_check;"
# Must return: ok
```

Step 7: Zero-Key Pre-Flight Self-Testing

1. Execute the platform's verification suites:

```
npm run test:structure
npm run test:data
npm run test:api
```

2. Verify root disk fail-closed protection:

```
node -e '
  const { checkRootDiskSpace } = require("./shared/lib/runtime/host_maintenance");
  checkRootDiskSpace("/");
  console.log("Root disk check passed: >= 1 GiB and >= 5% free.");
'
```

Step 8: Network Unfencing, Reverse Proxy Binding & Gateway Launch

1. Re-enable production network traffic:

```
iptables -P INPUT ACCEPT
iptables -F
```

2. Launch the auxiliary monitoring containers:

```
docker compose -f infra/docker/docker-compose.yml up -d
```

3. Start the Sovereign CLI background trading runner in dry-run monitoring mode:

```
node backend/cli/commands/runner/run.js --once
```

4. Confirm successful disaster recovery by polling the system health endpoint:

```
curl -s http://127.0.0.1:8787/health | jq .
# Expected source-level payload: { "ok": true, "service": "sovereign-web" }
# Cluster role is exposed and qualified separately; /health does not prove failover readiness.
```

Appendix B. Complete Platform Mathematical Invariants & Verification Matrix

The Sovereign Trading Platform enforces twelve fundamental mathematical invariants across all execution, storage, and analytics paths.

Table B.1: The 12 Platform Mathematical Invariants

ID	Invariant Name	Mathematical Formulation	Canonical Source Implementation	Verif Comı
INV-01	**Half-Open Intervals**	Temporal queries evaluate $[t_{\text{start}}, t_{\text{end}})$ such that $t \in [t_{\text{start}}, t_{\text{end}}) \iff t_{\text{start}} \le t < t_{\text{end}}$.	`shared/lib/market/ts_index_storage.js`	`npm test`
INV-02	**UTC Monotonic Primacy**	For any timeseries $\{B_k\}_{k=1}^{M}$, $t_k \ge t_{k-1}$ in UTC epoch milliseconds.	`shared/lib/data/quarantine.js`	`node scri ev/v _doc mula ty.j`
INV-03	**Exact Decimal Scaling**	Quantities and monetary amounts scale by 10^D in integer arithmetic before IEEE float conversion.	`shared/lib/strategy/backtest.js`	`npm test`
INV-04	**Bitemporal Point-In-Time**	$\text{Visible}(t) \iff t_{\text{publication}} \le T_{\text{decision}} \wedge t_{\text{receipt}} \le T_{\text{decision}}$.	`shared/lib/data/macro_store.js`	`test ipts grat esea nvar /mar nvar .tes`
INV-05	**Asymmetric Return Recovery**	$R_{\text{recovery}} = \frac{D}{1-D}$ where $D \in [0, 1)$ is the drawdown fraction.	`shared/lib/strategy/backtest.js`	`test ipts grat esea nvar /mar nvar .tes`
INV-06	**Available Cash Conservation**	$\text{AvailableCash}(t) = \max\left(0, \text{virtual_balance}(t) - \sum_{t_i > t} \text{receivable}_i\right) = \max(0, \text{settled_cash}(t))$.	`backend/gateway/src/polymarket/paper_ledger.ts`	`test ipts grat atew per_ r_se ent. js`
INV-07	**CLOB Spread Fallback**	$P_{\text{ref}} = \begin{cases} \frac{P_{\text{bid}} + P_{\text{ask}}}{2} & \text{if } P_{\text{ask}} - P_{\text{bid}} \le 0.10 \\ P_{\text{last}} & \text{if } P_{\text{ask}} - P_{\text{bid}} > 0.10 \end{cases}$.	`shared/lib/strategy/polymarket_clob_model.js`	`test ipts grat olym /pol et_c prea lbac t.js`
INV-08	**Harmonic Depth Imbalance**	$DIR_5 = \frac{\sum_{i=1}^{5} \frac{V_{b,i} - V_{a,i}}{i}}{\sum_{i=1}^{5} \frac{V_{b,i} + V_{a,i}}{i}} \in [-1, +1]$.	`shared/lib/strategy/polymarket_clob_model.js`	`node scri ev/v _doc mula ty.j`
INV-09	**Whalley-Wilmott Band**	$\Delta H = \left(\frac{3\lambda S_t \Gamma_{\text{net}}^2}{2\gamma}\right)^{1/3}$.	`shared/lib/strategy/options_smile_model.js`	`npm test`
INV-10	**SVI Arbitrage Absence (research target)**	$\frac{\partial w(k)}{\partial k} \in (-2, 2)$ and $g(k) \ge 0$ for all moneyness k.	No production owner yet; `options_smile_model.js` uses nearest-neighbor 25-	Not quali

ID	Invariant Name	Mathematical Formulation	Canonical Source Implementation	Verif Comm
			delta observations.	
INV-11	**Root Storage Fail-Closed**	$\text{Abort} \iff \text{FreeBytes} < 1024^3 \vee \text{FreePercent} < 5.0\%$.	`shared/lib/runtime/host_maintenance.js`	`test ipts atio ost_ enan st.j`
INV-12	**SHA-256 Forward Chaining**	$H_i = \text{SHA256}(\text{stableObject}(\text{event}_i))$ with $\text{prior_checksum} = H_{i-1}$.	`backend/gateway/src/polymarket/paper_ledger.ts`	`test ipts grat atew per_ r_se ent. js`

Appendix C. End-to-End System ASCII Architecture Schematics

Figure C.1: Master System Architecture & Micro-Component Dataflow

Figure C.1A: Master Architecture Portrait Overview

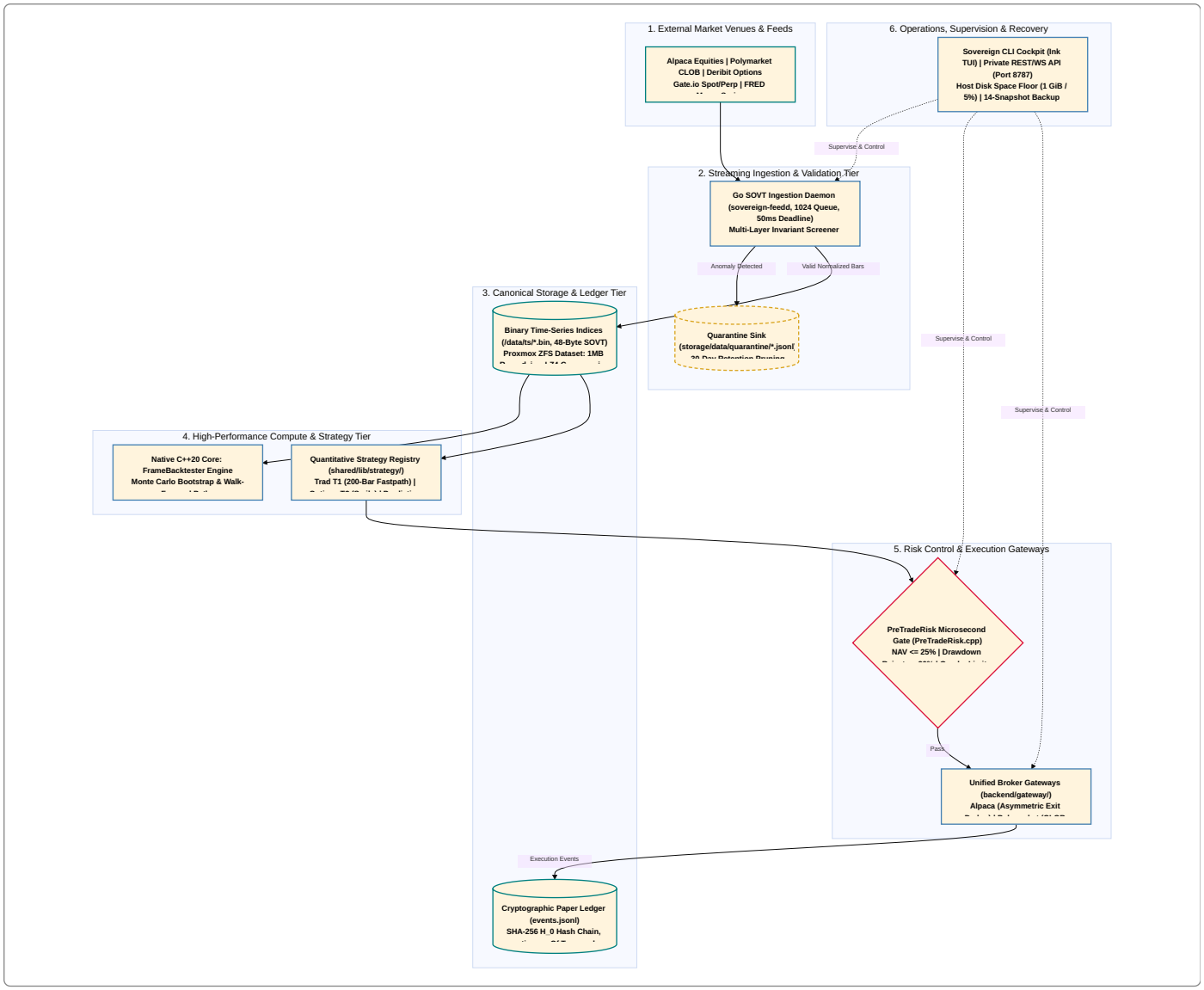


Figure C.1B: Panel 1 - Ingestion & Storage Pipeline

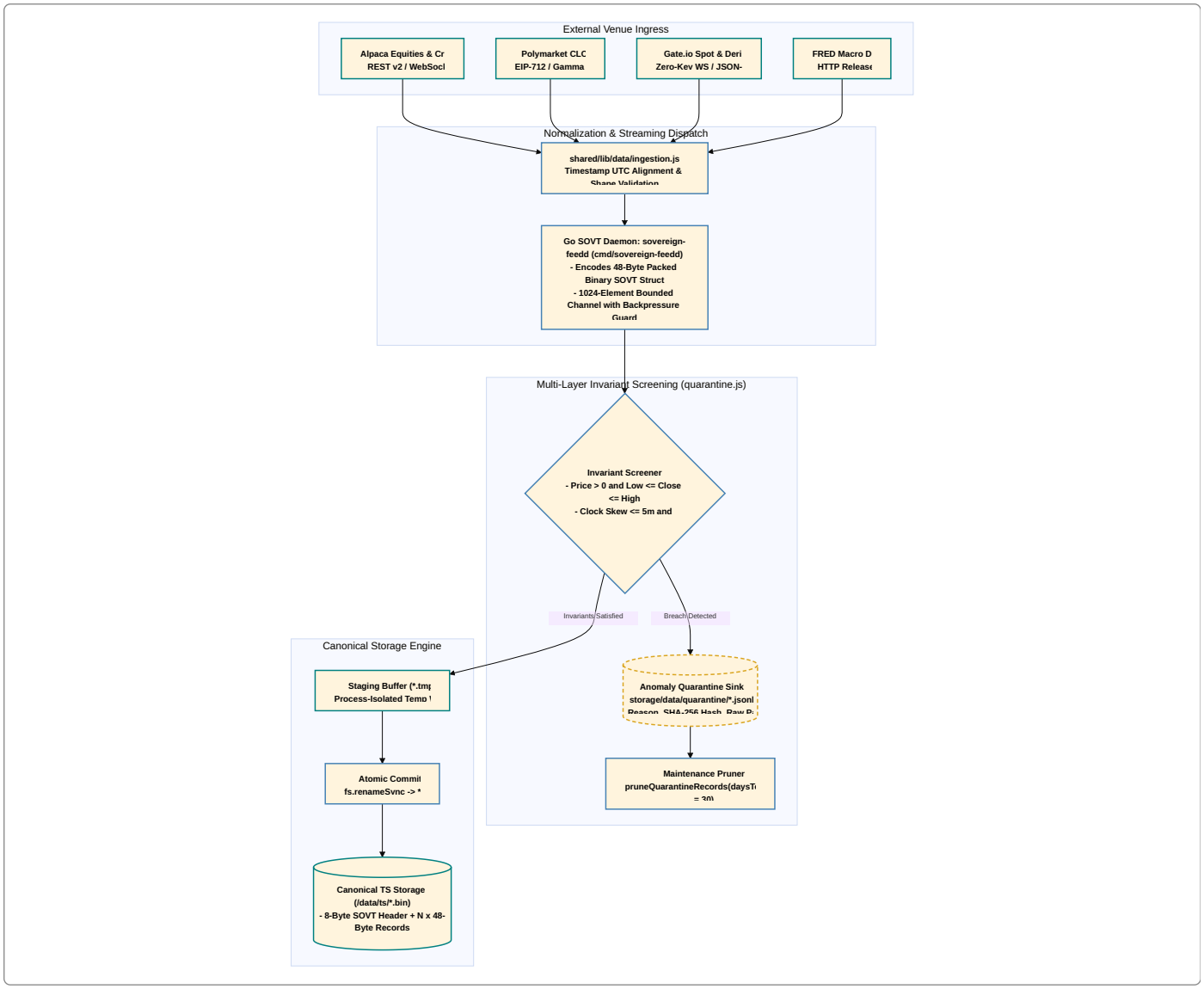


Figure C.1C: Panel 2 - Quantitative Strategy, Risk & Execution

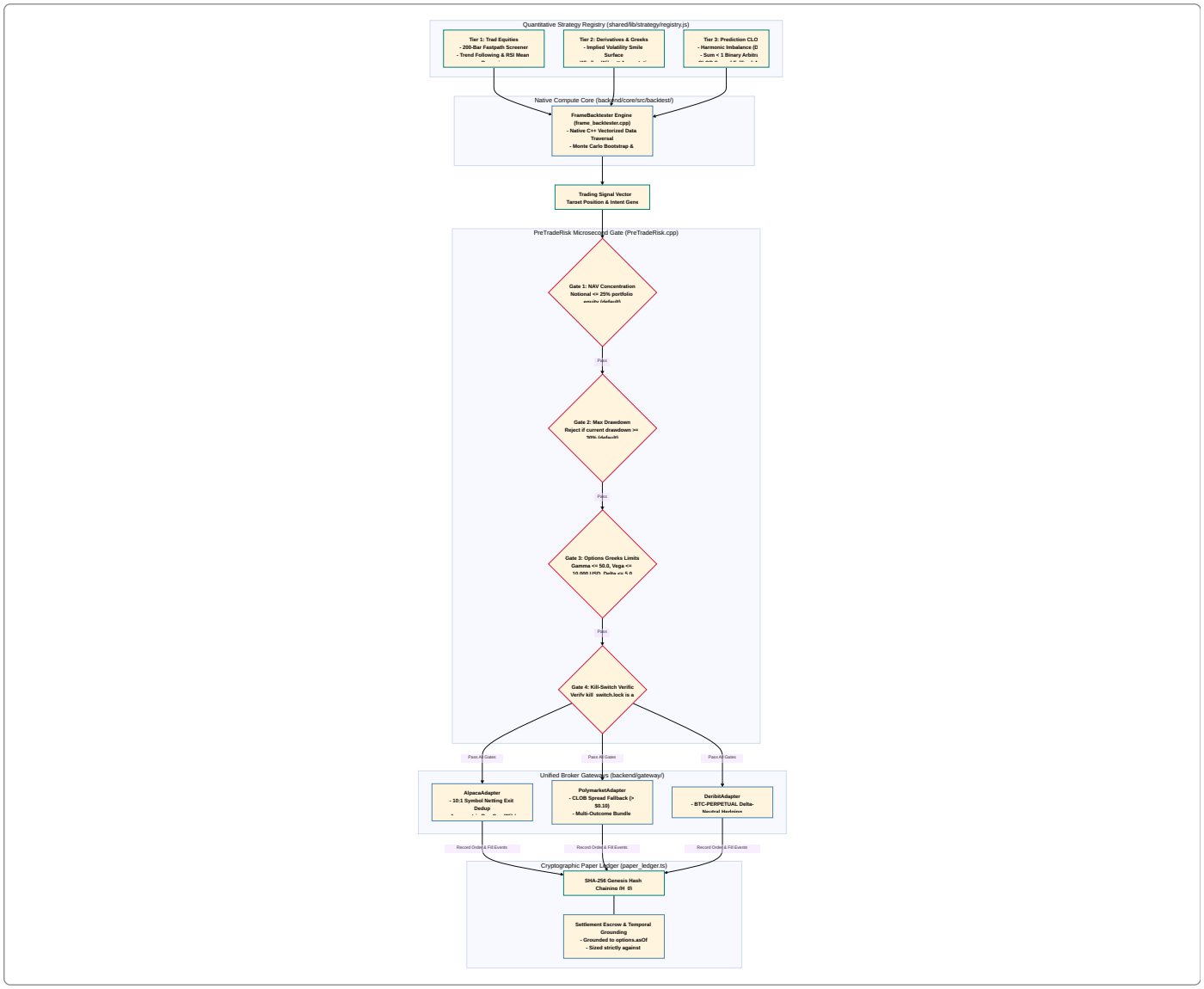
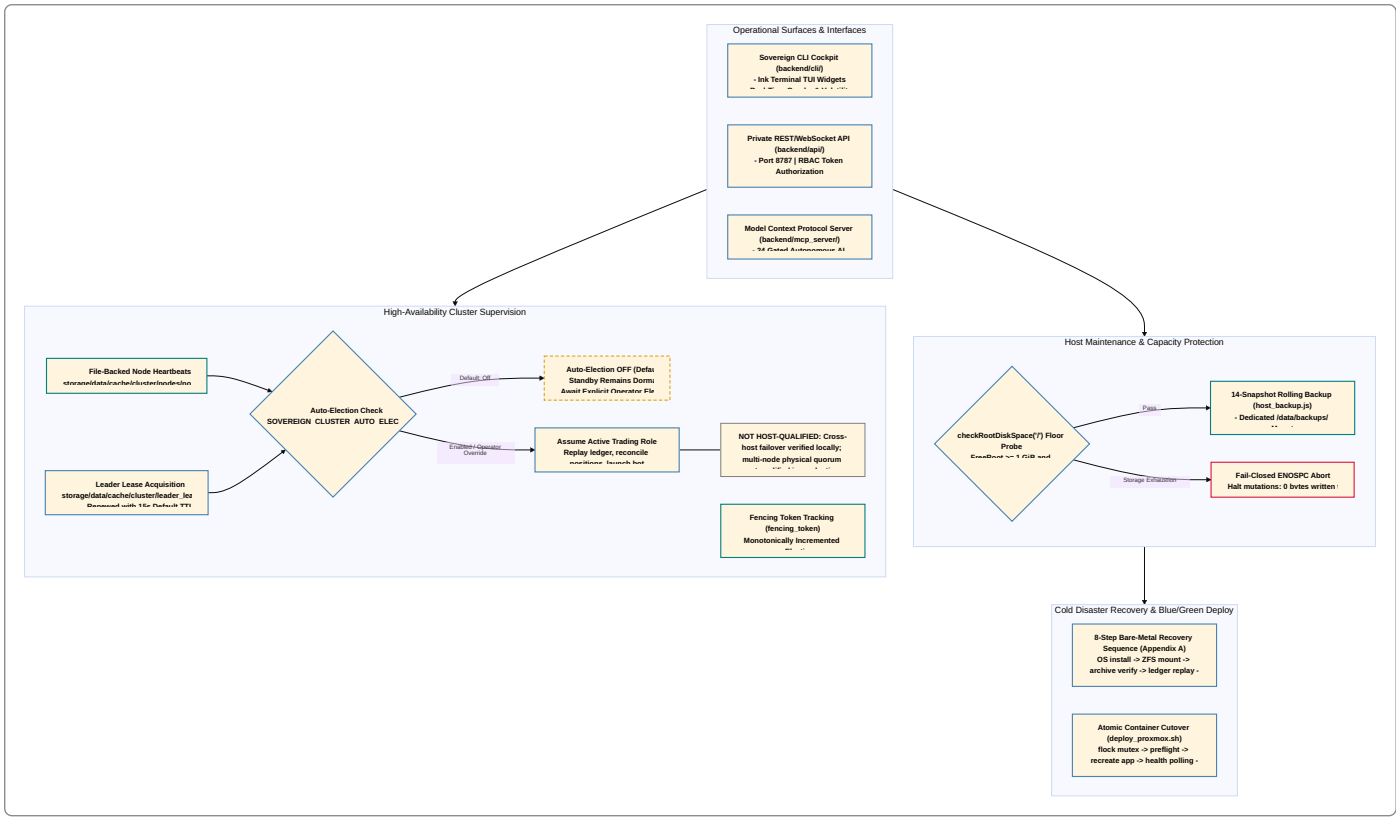


Figure C.1D: Panel 3 - Operations, Cluster & Disaster Recovery



Appendix D. Glossary of Quantitative & Systems Engineering Terms

- **AVX2 (Advanced Vector Extensions 2)**: An x86 instruction set architecture extension supporting 256-bit SIMD registers, enabling single-instruction processing of eight 32-bit floating-point values concurrently.
- **Bitemporal History**: A temporal data model distinguishing between *event time* (the real-world instant a transaction or economic observation occurred) and *system/publication time* (the exact instant the record was made available to the computing engine), preventing lookahead bias.
- **CLOB (Central Limit Order Book)**: A transparent exchange microstructure mechanism where continuous incoming buy and sell limit orders are matched deterministically based on price-time priority.
- **DIR5 (Depth Imbalance Ratio, Top-5)**: A quantitative order book metric that measures the normalized difference between total bid volume and ask volume across the top 5 levels weighted by the harmonic series $w_i = 1/i$.
- **EIP-712**: An Ethereum Improvement Proposal standard for hashing and signing typed structured data, enabling secure off-chain order placement without exposing private keys in plaintext.
- **Fail-Closed**: A system design principle ensuring that any unexpected failure, network timeout, corrupt record, or ambiguous state transition causes the platform to halt execution and reject new orders rather than proceeding under unvalidated assumptions.
- **LXC (Linux Containers)**: An operating system-level virtualization method for running multiple isolated Linux environments on a single host kernel using cgroups and kernel namespaces without hypervisor overhead.
- **NegRisk (Negative Risk Conversion)**: A smart contract mechanism deployed in multi-outcome prediction markets allowing traders to convert 1 unit of collateral into a complete set of N outcome tokens, or merge complementary tokens back into collateral.
- **OFI (Order Flow Imbalance)**: A high-frequency microstructure indicator that measures the net order volume driving price changes between consecutive order book snapshots.
- **O(1) Seeking Read**: A filesystem optimization technique that utilizes POSIX `lseek(2)` or `fs.readSync` to inspect the final fixed-size binary record of a file directly by jumping to `file_size - 48` bytes without parsing historical data.
- **PreTradeRisk**: An ultra-low latency risk validation engine executed in native compiled C++ that intercepts order intents and verifies portfolio Greek bounds, capital concentration limits, and kill-switch locks before network transmission.
- **RR25 (25-Delta Risk Reversal)**: The difference in implied volatility between a 25-delta call option and a 25-delta put option, quantifying market skew and directional hedging premium.
- **SIMD (Single Instruction, Multiple Data)**: A parallel computing technique that allows a CPU processor to perform the same arithmetic operation on multiple data elements simultaneously using vector registers.
- **SOVT (Sovereign Open Vector Time-Series)**: The platform's canonical binary time-series format comprising an 8-byte header (`SOVT` magic + `uint32LE` count) followed by contiguous, unpadded 48-byte records (`6 × float64LE`).
- **SVI (Stochastic Volatility Inspired)**: A five-parameter implied-variance surface family introduced by Gatheral. Parameterization alone does not guarantee static-arbitrage absence; calibrated constraints must be checked explicitly. SVI is a research target in this repository.
- **VWAP (Volume-Weighted Average Price)**: The average execution price of a trade weighted by the volume traded at each distinct price level across an order book depth ladder.

- **WAF (Write Amplification Factor)**: The numerical ratio of physical flash bytes written to storage media relative to the logical bytes requested by the operating system, quantifying SSD flash cell wear.
- **Whalley-Wilmott Model**: An asymptotic optimal control formulation for delta-hedging under proportional transaction costs that establishes dynamic no-trade bands around target hedge ratios.
- **ZFS Recordsize**: The maximum block allocation size for files stored within a ZFS dataset, optimized to 1MB for sequential time-series scanning.

References

1. **3GPP TS 23.003 & GSMA TS.06:** Structural identification and checksum standards.
2. **IEEE 754-2019:** Floating-point arithmetic (double precision, little-endian).
3. **ISO/IEC 14882:2020:** Programming Languages — C++ (C++20 Standard).
4. **POSIX.1-2017 (IEEE Std 1003.1-2017):** Portable Operating System Interface (`statfs`, `flock`, `rename`).
5. **EIP-712:** Ethereum Structured Data Hashing and Signing (Polymarket CLOB auth).
6. **Node.js v20 LTS Documentation:** Native Filesystem, Streams, Crypto, and Test Runner.
7. **Gatheral, J. (2004):** *A Parsimonious Model of the Volatility Surface*, Presentation at Quantitative Methods in Finance.
8. **Whalley, A. E., & Wilmott, P. (1997):** *An Asymptotic Analysis of an Optimal Hedging Model for Option Portfolios with Transaction Costs*, Mathematical Finance, 7(3), 307-324.
9. **Sovereign Repository Sources:** `backend/core/`, `shared/lib/`, `backend/gateway/`, `infra/docker/`, `docs/Reports/`.